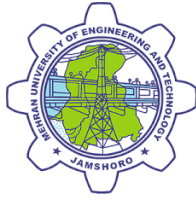


ROLL NO:



MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY
JAMSHORO



ELECTRONIC CIRCUIT DESIGN

Lab Manual

BATCH-18ES

SEMESTER-3RD

YEAR -2ND

DEPARTMENT OF ELECTRONIC ENGINEERING

PREPARED BY:
Engr. QUDSIA MEMON



Department of Electronic Engineering

Certificate

Certified that Mr. /Miss. _____

S/O/D/O _____ of 3th Semester 2nd Year of
Electronic Engineering has carried out the necessary practical work for
the subject of ELECTRONIC CIRCUIT DESIGN as prescribed by the
Mehran University of Engineering & Technology, Jamshoro.

Engr. Qudsia Memon
Lecturer

COURSE NAME: ELECTRONIC CIRCUIT DESIGN- ES-203 (PRACTICAL)

LAB PERFORMANCE INDICATOR	SUBJECT KNOWLEDGE	DATA ANALYSIS AND INTERPRETATION	CALCULATION AND CODING	ABILITY TO CONDUCT EXPERIMENT	OBSERVATION/ RESULTS	PRESENTATION/ REPORT	SCORE
OBJECTIVE NO:							
01	Introduction to Electronic Circuit Design involving Multisim Simulation tool						
02	BJT Small Signal Common Emitter Amplifier						
03	BJT Small Signal Common Collector Amplifier						
04	BJT Small Signal Common Base Amplifier						
05	FET Small Signal Common Source Amplifier						
06	To observe the Frequency Response of Common Emitter Amplifier						
07	To observe the Frequency Response of Common Source Amplifier						
08	To observe the Operating Characteristics of a Push pull amplifier						
09	To observe the Operating Characteristics of a Tuned Amplifier						
10	To Understand the Operation of R-C Coupled Two Stage Bipolar Junction Transistor Amplifiers						
11	To Understand the Operation of Colpitts Oscillator						
12	To Understand the Operation of Hartley Oscillator						
13	To Understand the Operation of RC-Phase Shift Oscillator						
14	To understand the operation of Wien-Bridge oscillators						
15	Open Ended Lab						
TOTAL MARKS OUT OF 5							

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 1

Name: _____ Roll No: _____

Score: _____ Signature of the Lab Tutor _____ Date: _____

LAB PERFORMANCE INDICATOR	Subject knowledge	Ability to conduct experiment	Observation/results	Presentation/ Report	Score
TOTAL	2	1		2	5
OBTAINED					

Introduction to Electronic Circuit Design involving Multisim Simulation tool

Learning Outcomes:

At the conclusion of this experiment, understudy will be able to

- Introduce the concept of software designing relative to hardware.
- Repeat the given circuits using Multisim.
- Imitate the given electronic circuits with a small change of parameters.

Equipment:

PC with Multisim installed.

Theory:

Electronic Circuit Design:

An each crucial electronic gadget built as an individual unit. Prior to the advancement of computerized circuits (ICs), every single individual transistor, diodes, resistors, capacitors, and inductors were discrete in nature. Any circuit or a framework can deliver the favored yield dependent on its input.

The procedure of circuit design can cover frameworks going from complex electronic frameworks right down to the individual transistors inside an incorporated circuit. For basic circuits, the structure procedure should frequently be possible by one individual without requiring an arranged or organized plan procedure and the planner ought to have the option to anticipate the voltage and current at each node in the circuit.

Simulation Software & Breadboard:

So as to briefly develop a circuit without harming the segments used to manufacture it, we should have a type of a stage that will both hold the segments set up and give the required electrical associations. Though progressively complex methods and gadgets have been created to make the get together and testing of electronic circuits simpler, but the idea of the breadboard still stays to accumulate segments on a short term stage.

The electronic circuit test system utilizes numerical models to recreate the conduct of a genuine electronic circuit. Simulation software considers demonstrating of circuit activity and is a significant examination device. As a result of the restriction of the breadboard and costly apparatuses like photomasks for integrated circuits, the majority of the IC configuration depends on simulation.

Introduction to Multisim:

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 1

NI Multisim is an incredible schematic catch and simulation environment that engineers, understudies, and teachers can use to mimic electronic circuits and model Printed Circuit Boards (PCBs).

Procedure:

1. How to Start Multisim:

For Windows users the default location can be found by clicking: Go to Start→Program→Multisim and click on Multisim.

You should see a screen similar to Figure 1 below. This is called as a “Capture and simulate” environment because you “Capture” your schematic by drawing it in MultiSim and then you “Simulate” it. In Fig. 1 important toolbars and menu are also labeled.

Open/Create Schematic

A blank schematic Circuit 1 is automatically created. To create a new schematic click on File – New – Schematic Capture. To save the schematic click on **File /Save As**. To open an existing file click on **File/ Open** in the toolbar.

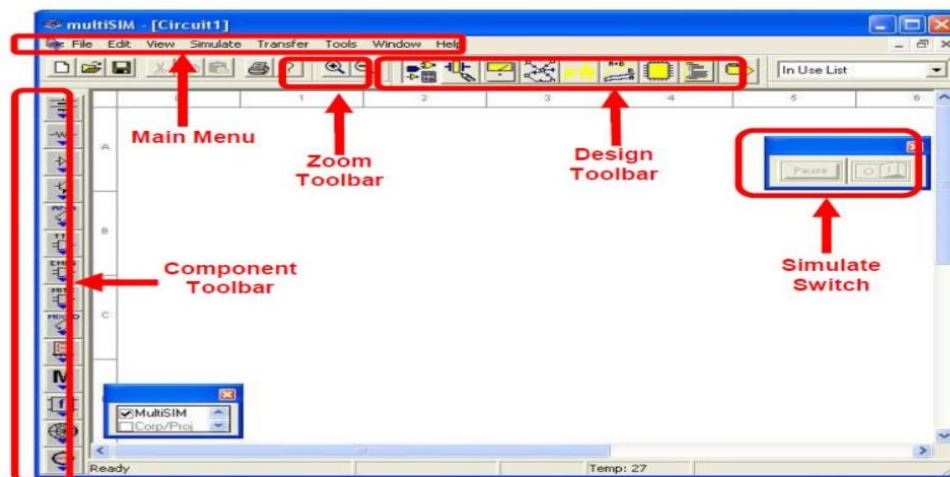


Fig. 1 Main Window of Multisim Simulation Software

2. Place Components:

To Place Components click on **Place/Components**. On the Select Component Window click on **Group** to select the components needed for the circuit. Click OK to place the component on the schematic.

For example to select resistors and the DC source click on **Place/ Components**. In **Group** select **Basic** scroll down to Resistors and select the value of the resistor needed to construct the circuit, let's say select 1k. To place DC source click on **Sources** in **Group** and select DC Source. As shown in Figure 2 and Figure 3 respectively.

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 1

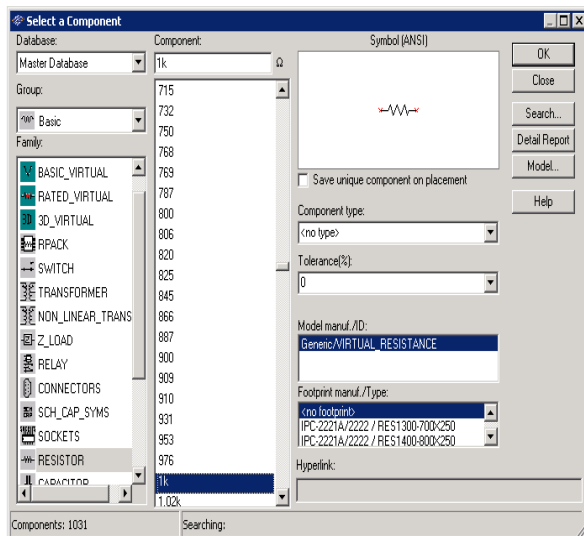


Figure 2: Select Resistor

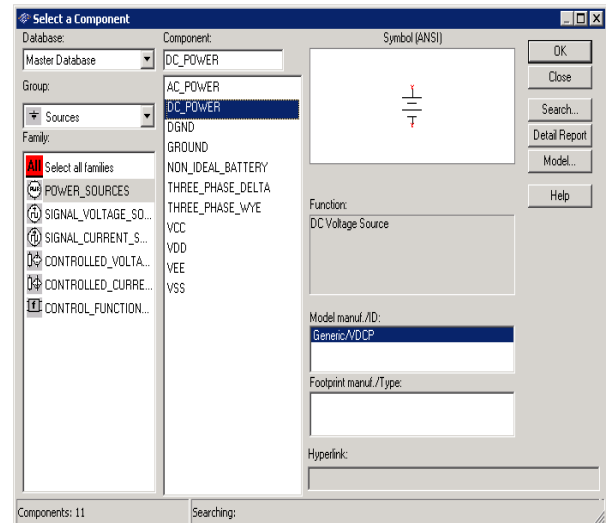


Figure 3: Select DC voltage

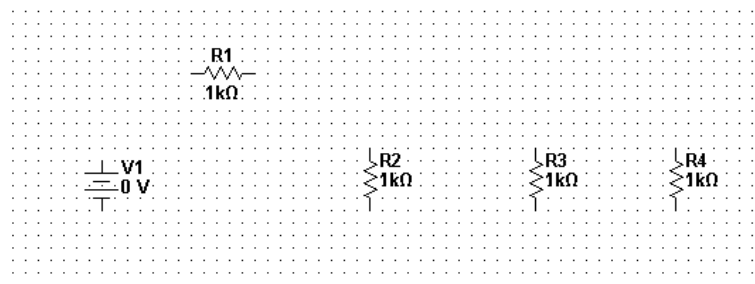


Figure 4: DC Source & Resistors

3. Rotate Components:

To rotate the components right click on the Resistor to flip the component on 90 Clockwise (Ctrl +R) and 90 Counter Clockwise (Ctrl+Shift+R).

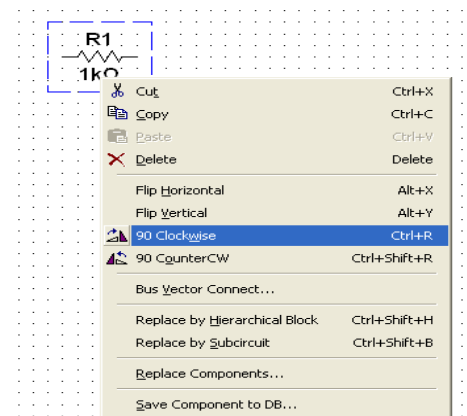


Figure 5: Rotate Components

4. Place Wire/Connect Components:

To connect resistors, click on **Place/Wire** drag and place the wire. Components can also be connected by clicking the mouse over the terminal edge of one component and dragging to the edge of another component. See Figure 6 for reference.

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 1

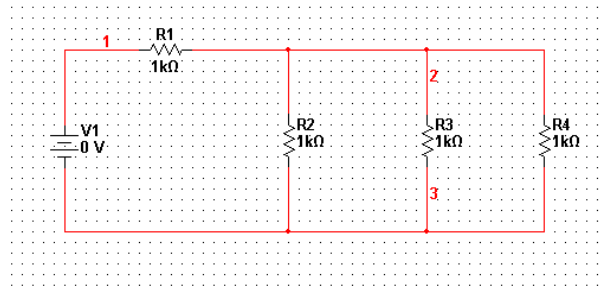


Figure 6: Place/ Wire

5. Change Component Values:

To change component values double click on the component this brings up a window that display the properties of the component. See Figure 7 for reference. Change R1 from 1k Ohm to 10 Ohms, R2 to 20 Ohms, R3 to 30 Ohms, and R4 to 40 Ohms. Also change the DV source from 0 V to 20 V.

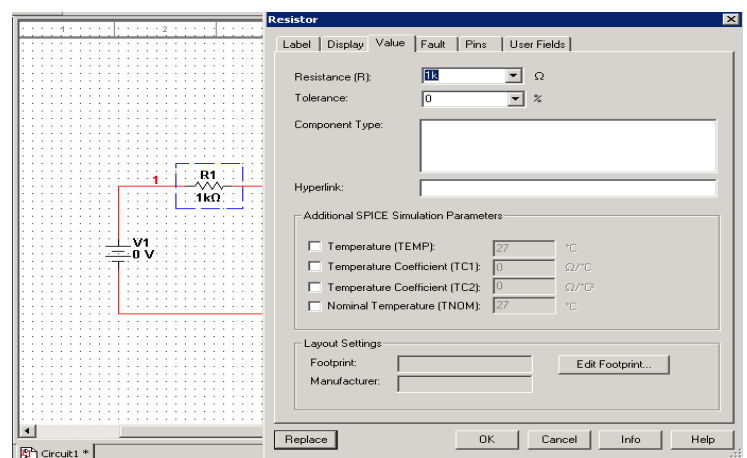


Figure 7: Change Component Values

6. Grounding:

All circuits must be grounded before the circuit can be simulated. Click on the Ground in the toolbar to ground the circuit. If the circuit is not grounded Multisim will not run the simulation.



Figure 8: Grounding

7. Simulation:

To simulate the completed circuit, Click on **Simulate/Run** or **F5**. This feature can also be accessed from the toolbar as shown in the Figure 9 below.



Figure 9: Simulation

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 1

Analyzing Components:

Multisim offers multiple ways to analyze the circuit using virtual instruments. Some of the basic instruments needed for this lab are described below.

1) Voltmeter:

The Voltmeter offers advantages over the multimeter for measuring voltage in a circuit.

Always connect the voltmeter in parallel with the load. The voltmeter can be found in the measurement toolbar.

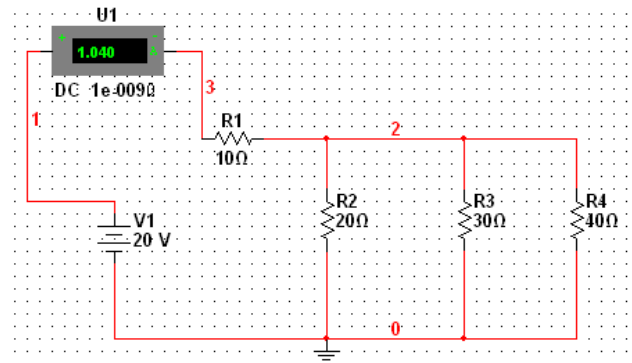


Figure 10: Voltmeter

2) Ammeter:

The ammeter offers advantages over the multimeter for measuring current in a circuit. It takes up less space in a circuit and you can rotate its terminals to suit your layout. **Always connect the ammeter in series with the load.** To place **Ammeter** click on View--- Toolbar --- Select Measurement Components. See Figure 11 on how to use the Ammeter.

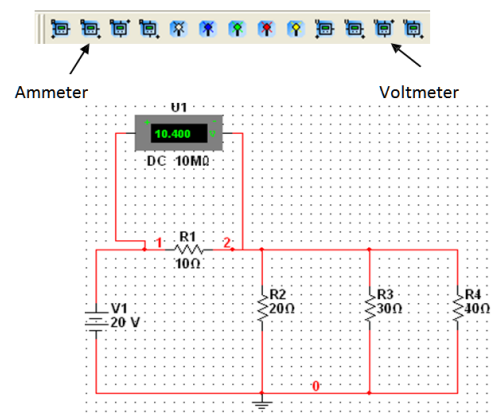


Figure 11: Ammeter

3) Multimeter:

Use the Multimeter to measure AC or DC voltage or current, and resistance or decibel loss between two nodes in a circuit. To use the Multimeter click on the Multimeter button in the **Instruments** toolbar and click to place its icon on the workspace. Double-click on the icon to open the instrument face, which is used to enter settings and view measurements.

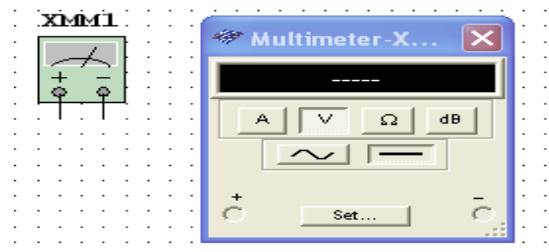


Figure 12: Multimeter

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 1

To measure Voltage place multimeter in Parallel with the component (Resistor, Voltage etc).
To measure Current place the multimeter in series with the component. Reference the Figure 13 and 14.

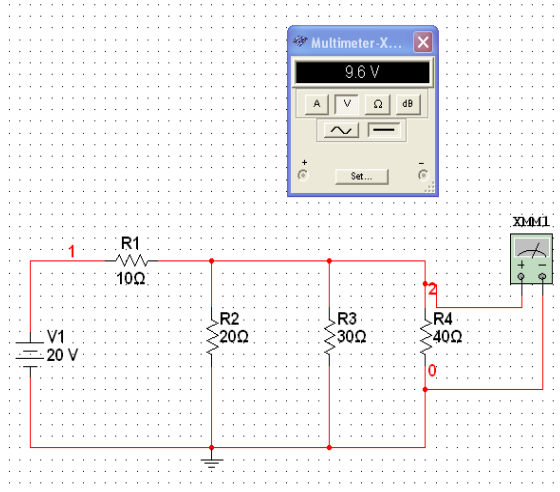


Figure 13: Measure Voltage

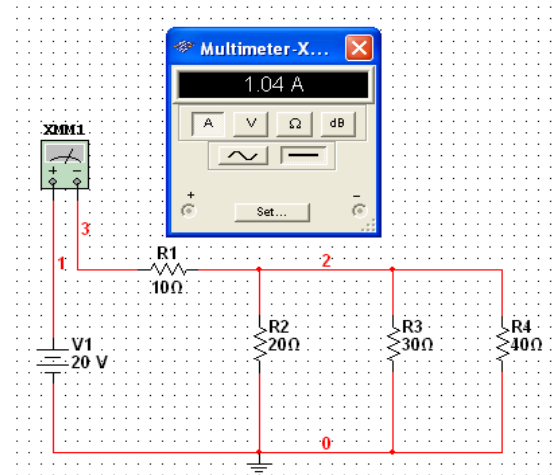


Figure 14: Measure Current

Tasks:

- Build a circuit comprising of 3 resistors (of any value) in series and a 15 V battery with multimeters across each resistor to calculate voltage across each.
- Build another circuit, utilizing two resistors, one of 200Ω resistance & another of 1KΩ resistance, a SPST (Single Pole Single Throw) switch, Direct Current Power Source, & a Ground connection. Afterwards, attach multimeter to measure the voltage output near the 1 KΩ resistor.

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 2

Name: _____ Roll No: _____

Score: _____ Signature of the Lab Tutor _____ Date: _____

LAB PERFORMANCE INDICATOR	Subject knowledge	Ability to conduct experiment	Observation/results	Presentation/ Report	Score
TOTAL	1	2	1	1	5
OBTAINED					

BJT Small Signal Common Emitter Amplifier

Learning Outcomes:

At the conclusion of this experiment, understudy will be able to

- Introduce the concept of small signal amplifiers.
- Determine the operation of a single stage Bipolar Junction transistor when used as a common emitter (CE) small signal amplifier.
- Replicate the given CE circuit using kit and breadboard.
- Imitate the CE amplifier circuit design with a small change of parameters.

Required Equipment and Components:

Equipment	Components
Oscilloscope	Bipolar Junction Transistor (BJT), type 2N3904
General-purpose function generator	Fixed resistor (68k Ω , 27k Ω , 1.2K Ω , 680 Ω)
0 – ± 15 Power supply	2 Capacitors (100 μ F)
Feed Back Trainer kit	
Breadboard	
Digital Multi-meter	

Theory:

Amplifier is the conventional term used to portray a circuit which delivers and expanded variant of its input signal. Be that as it may, not all amplifier circuits are equivalent to they are grouped by their circuit arrangements and methods of activity.

In “Electronics”, small signal amplifiers are normally utilized tools as they can intensify a generally little input signal, for instance from a Sensor such as a photo-device, into an a lot bigger output signal to drive a relay, lamp or loudspeaker for instance.

Small Signal Amplifiers

- Small Signal Amplifiers are otherwise called Voltage Amplifiers.
- Voltage Amplifiers have 3 fundamental properties, Input Resistance, Output Resistance and Gain.
- The Gain of a small signal amplifier is the amount by which the amplifier “Amplifies” the input signals.
- In order to amplify all of the input signal distortion free in a Class A type amplifier, DC Base Biasing is required.

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 2

- DC Bias sets the Q-point of the amplifier half way along the load line.
- This DC Base biasing implies that the amplifier devours power regardless of whether there is no input signal present.
- The transistor amplifier is non-linear and an incorrect bias setting will produce large amounts of distortion to the output waveform.
- Too large an input signal will produce large amounts of distortion due to clipping, which is also a form of amplitude distortion.
- Incorrect positioning of the Q-point on the load line will produce either Saturation Clipping or Cut-off Clipping.

The Common Emitter Amplifier arrangement is the most widely recognized type of all the universally useful voltage amplifier circuit utilizing a Bipolar Junction Transistor.

Mode of Operation:

Fig. 1 shows the circuit of a single-stage CE amplifier using an NPN transistor. Here, base is the driven element. The input signal is injected into the base emitter circuit whereas output signal is taken out from the collector emitter circuit. The E/B junction is forward biased by V_{BB} and C/B junction is reversed-biased by V_{CC} (in fact, same battery V_{CC} can provide dc power for both base and collector). The Q-point or working condition is determined by V_{CC} together with R_B and R_C .

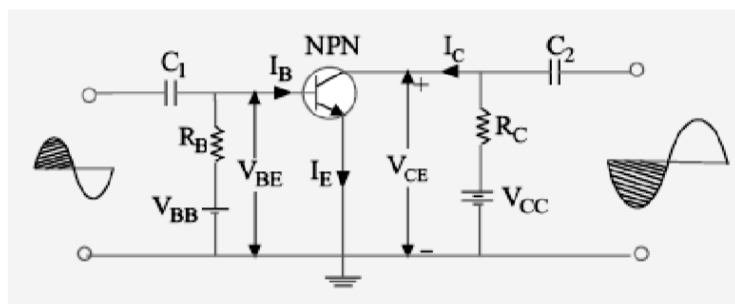


Fig. 1

The single stage common emitter amplifier circuit shown below uses what is commonly called “Voltage Divider Biasing”. This sort of biasing arrangement utilizes two resistors as a potential divider network across the supply with their center point supplying the required Base bias voltage to the transistor. Voltage divider biasing is commonly used in the design of bipolar transistor amplifier circuits.

This method of biasing the transistor greatly reduces the effects of varying Beta, (β) by holding the Base bias at a constant steady voltage level allowing for best stability.

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 2

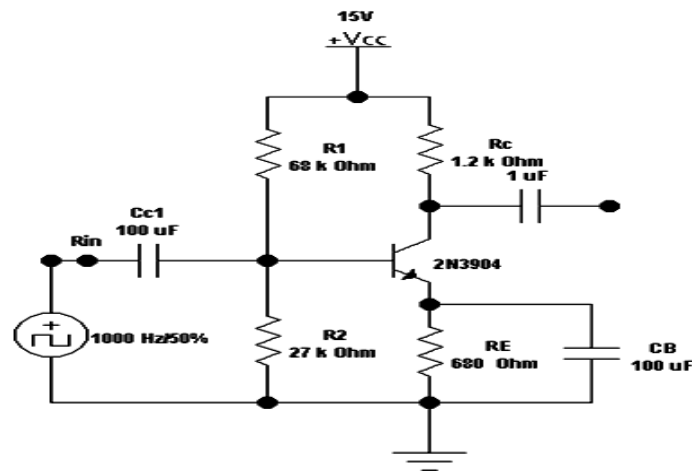


Figure 2: Common Emitter Amplifier

Characteristics of a CE Amplifier:

A CE transistor amplifier has the following characteristics:

1. It has moderately low input resistance (1 K to 2 K),
2. Its output resistance is moderately large (50 K or so),
3. Its current gain (β) is high (50–300),
4. It has very high voltage gain of the order of 1500 or so,
5. It produces very high power gain of the order of 10,000 times or 40 dB,
6. It produces phase reversal of input signal i.e. input and output signals are 180° out of phase with each other.

Procedure:

1. Utilizing the kit, make the connections according to Figure 2 using the Bypass circuit that is capacitor C_2 should be connected through emitter resistor R_E .
2. Give an input to the amplifier so that the output is 1 Vpp at 1 kHz Measure the input voltage ($f=1$ kHz).
3. Adjust the input signal through function generator until the output sine wave is no longer clipped which is being displayed through oscilloscope.
4. Observe voltages and currents at different nodes.
5. Calculate the voltage gain of the amplifier by observing V_i and V_o .
6. Using A_v , determine the A_i and A_p .
7. Now remove the Bypass circuit that is capacitor C_2 should be unconnected, repeat the same procedure and compare the results.
8. Replicate steps 1-7 using bread board.

Observation/Results:

CE configuration:

Transistor code: _____, Transistor type: _____ (PNP/NPN) $R_B =$ _____, $R_C =$ _____
 $R_E =$ _____, $C_{C1} =$ _____, $C_B =$ _____

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 2

With Bypass Capacitor:

Source	V_i	V_o	I_i	I_o	Z_{in}	Z_{out}	A_v	A_i	A_p	Power gain (dB)
Kit										
Bead Board										

Without Bypass Capacitor:

Source	V_i	V_o	I_i	I_o	Z_{in}	Z_{out}	A_v	A_i	A_p	Power gain (dB)
Kit										
Bead Board										

Review Questions:

- What effect does By-pass capacitor have on the voltage gain?

- Capacitors C1 and C2 are used as Coupling Capacitors, Why are they used?

- Why capacitors don't take part in DC analysis?

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 2

- All small signal amplifiers work in an active region. But to design these amplifier circuits is it necessary to draw a "load line chart"? If not then in order to design a BJT amplifier what parameters are required?

- Why we need to know Input resistance to an amplifier?

Task:

Replicate the CE amplifier circuit in multisim by varying the values of R_1 , R_2 , R_E , R_C , C_{C1} one by one and explain the effect of changing the value of a given component respectively.

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 2

Calculation Formulas:

a) Voltage Gain (A_v):

- Voltage gain is the ratio of output voltage to input voltage. $A_v = \frac{V_o}{V_i}$
- Observe input ac voltage V_i at the terminal of Coupling Capacitor1.
- Observe output ac voltage V_o at the collector.

b) Current Gain (A_i):

- Current gain is the ratio of collector current (I_o) to the base current (I_i). $A_i = \frac{I_o}{I_i}$
- For input current I_i , according to ohm's law. $I_i = V_i / (R_1 \parallel R_2)$
- For output current I_o . $I_o = \frac{V_o}{R_c}$

c) Power Gain (A_p):

- It is the ratio of output signal power to input signal power.
- Power Gain = $A_p = A_v \times A_i$
- Power gain in $dB = 10 \log A_p$

d) Input Resistance:

Small signal input resistance is measured at the input of the transistor to the small signal ground.

$$Z_{in} = \frac{V_{in}}{I_{in}}$$

e) Output Resistance:

Small signal output impedance Z_{out} can be measured by removing (short circuiting) the input voltage source V_s and inserting a test voltage source between the output terminals (parallel to R_E).

$$Z_{out} = \frac{V_{out}}{I_{out}}$$

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 3

Name: _____ Roll No: _____

Score: _____ Signature of the Lab Tutor _____ Date: _____

LAB PERFORMANCE INDICATOR	Subject knowledge	Ability to conduct experiment	Observation/results	Presentation/ Report	Score
TOTAL	2	1	1	1	5
OBTAINED					

BJT Small Signal Common Collector Amplifier

Learning Outcomes:

At the conclusion of this experiment, understudy will be able to

- Determine the operation of a single stage Bipolar Junction transistor when used as a common collector (CC) small signal amplifier.
- Replicate the given CC circuit using kit and breadboard.
- Imitate the CC amplifier circuit design with a small change of parameters.

Required Equipment and Components:

Equipment	Components
Oscilloscope	Bipolar Junction Transistor (BJT), type 2N3904
General-purpose function generator	2 Capacitors ($100\mu\text{F}$)
0 – +15 Power supply	Fixed resistor ($68\text{k}\Omega$, $27\text{k}\Omega$, 1.2Ω , 680Ω)
Feed Back Trainer kit	
Breadboard	
Digital Multi-meter	

Theory:

From numerous point of views the common collector configuration (CC) is the reverse of the common emitter (CE) configuration as the connected load resistor is changed from the collector terminal for R_C to the emitter terminal for R_E .

The common collector or grounded collector configuration is commonly used where a high impedance input source needs to be connected to a low impedance output load requiring a high current gain. The former conditions make CC an ideal buffer between a high impedance source and a low impedance load. A buffer is any circuit that keeps the source from being affected by a load.

Mode of Operation:

Fig.1 and 2 shows the circuit of a single-stage CC amplifier using an NPN transistor. The input signal is injected into the base-collector circuit and output signal is taken out from the emitter collector circuit. The E/B junction is forward-biased by VEE and C/B junction is reverse-biased by VCC. The quiescent values of I_B and I_E are set by VCC and VEE together with R_B and R_E .

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 3

$$I_E = \frac{V_{EE} - V_{BE}}{R_E + R_B / \beta}$$

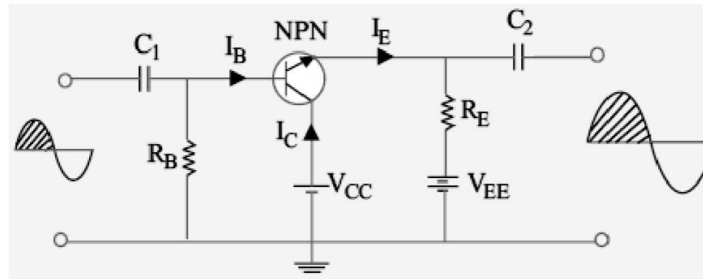


Fig. 1

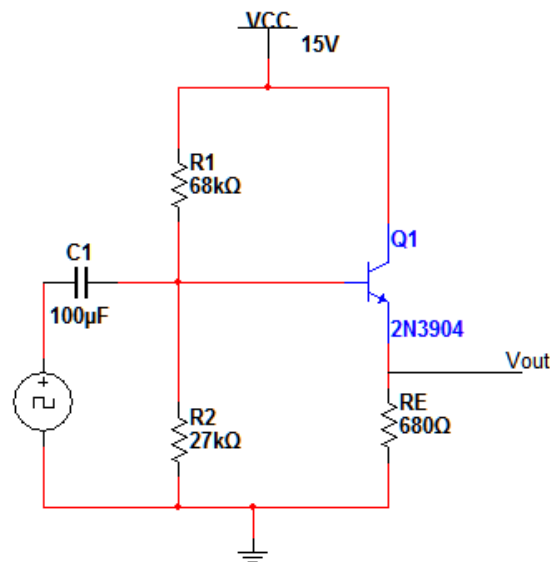


Figure 2: Common Collector Amplifier

Characteristics of a CC Amplifier:

A CC amplifier has the following characteristics:

1. High input impedance (20-500 K),
2. Low output impedance (50-1000 Ω),
3. High current gain of $(1 + \beta)$ i.e. 50 – 500,
4. Voltage gain of less than 1,
5. Power gain of 10 to 20 dB,
6. No phase reversal of the input signal.

Procedure:

1. Utilizing the kit, make the connections according to Figure 2 without using the Bypass circuit that is capacitor C_2 should remains unconnected.
2. Give an input to the amplifier so that the output is 1 Vpp at 1 kHz Measure the input voltage ($f=1$ kHz).
3. Adjust the input signal through function generator until the output sine wave is no longer clipped which is being displayed through oscilloscope.

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 3

4. Observe voltages and currents at different nodes.
5. Then calculate the voltage gain of the amplifier.

Observation/Results:

CC configuration:

Transistor code: _____, Transistor type: _____ (PNP/NPN) $R_B =$ _____, $R_E =$ _____
 $R_C =$ _____, $C_{C1} =$ _____, $C_B =$ _____

Source	V_i	V_o	I_i	I_o	Z_{in}	Z_{out}	A_v	A_i	A_p	Power gain (dB)
Kit										
Bread Board										

Review Questions:

- Another name for the CC amplifier is the emitter follower. Based on the gain that you have found, why do you think it is called that?

- Why CC amplifier is useful for impedance matching?

- Why the output signal present at the emitter is in-phase with the applied input signal at the base?

- Which operating mode is used when the transistor is being utilized as an amplifier? What is the condition or desired biasing of junctions for that operating mode? Does

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 3

that condition changes, when the transistor is configured as CC? Briefly explain your answer in terms of biasing.

- Does the voltage gain vary, by varying input voltage? Also, give reasons.

- Does varying input voltage effect the current gain? Also, give reasons.

Task:

Replicate the CC amplifier circuit in multisim by varying the values of R_1 , R_2 , R_E , R_C , C_{C1} one by one and explain the effect of changing the value of a given component respectively.

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 4

Name: _____ Roll No: _____

Score: _____ Signature of the Lab Tutor _____ Date: _____

LAB PERFORMANCE INDICATOR	Subject knowledge	Ability to conduct experiment	Observation/results	Presentation/ Report	Score
TOTAL	2	1	1	1	5
OBTAINED					

BJT Small Signal Common Base Amplifier

Learning Outcomes:

At the conclusion of this experiment, understudy will be able to

- Determine the operation of a single stage Bipolar Junction transistor when used as a common base (CB) small signal amplifier.
- Replicate the given CB circuit using kit and breadboard.
- Imitate the CB amplifier circuit design with a small change of parameters.

Required Equipment and Components:

Equipment	Components
Oscilloscope	Bipolar Junction Transistor (BJT), type 2N3904
General-purpose function generator	Few resistors
0 – ± 15 Power supply	2 Capacitors
Feed Back Trainer kit	
Breadboard	
Digital Multi-meter	

Theory:

The Common Base Amplifier is another type of bipolar junction transistor, (BJT) configuration where the base terminal of the transistor is a common terminal to both the input and output signals, hence its name common base (CB). The common base configuration is less common as an amplifier than compared to the more popular common emitter, (CE) or common collector, (CC) configurations but is still used due to its unique input/output characteristics.

The common-base amplifier can provide a reasonable level of voltage gain but suffers from low input impedance and a current gain of less than one. However, this circuit is used in high-frequency applications because its terminal characteristics at high frequencies are better than those of a common-emitter configuration using the same transistor. The low input impedance of the common base amplifier will limit its use to specialized RF applications.

Mode of Operation:

Fig. 1 shows the circuit of a single-stage CB amplifier using NPN transistor. As seen, input ac signal is injected into the emitter-base circuit and output is taken from the collector-base circuit. The E/B junction is forward-biased by VEE where as C/B junction is reverse-biased

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 4

by V_{CC} . The Q-point or dc working conditions are determined by dc batteries along with resistors R_E and R_C . In other words, values of I_E , I_B and V_{CB} are decided by V_{CC} , V_{EE} , R_E and R_C . The voltage V_{CB} is given by the equation $V_{CB} = V_{CC} - I_C R_C$. When no signal is applied to the input circuit, the output just sits at the Q-point so that there is no output signal.

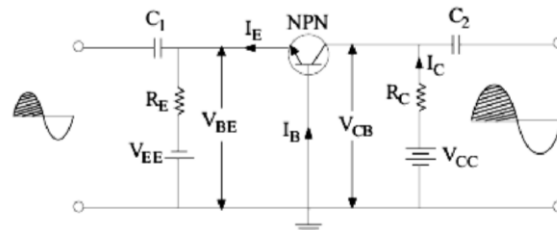


Fig. 1

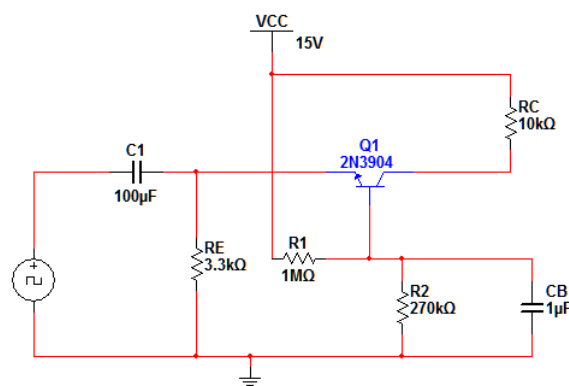


Figure 2: Common Base Amplifier

Characteristics of a CB Amplifier:

A CB amplifier has the following characteristics:

1. Very low input resistance ($30 - 150 \Omega$).
2. Very high output resistance (up to 500 K).
3. A current gain $\alpha < 1$.
4. Large voltage gain of about 1500.
5. Power gain of up to 30 dB.
6. No phase reversal between input and output voltages.

Procedure:

1. Utilizing the kit, make the connections according to Figure 2 without using the Bypass circuit that is capacitor C_2 should remains unconnected.
2. Give an input to the amplifier so that the output is 1 V_{pp} at 1 kHz Measure the input voltage ($f=1$ kHz).
3. Adjust the input signal through function generator until the output sine wave is no longer clipped which is being displayed through oscilloscope.
4. Observe voltages and currents at different nodes.
5. Then calculate the voltage gain of the amplifier.

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 4

Observation/Results:

CB configuration:

Transistor code: _____, Transistor type: _____ (PNP/NPN) $R_E =$ _____, $R_C =$ _____
 $R_B =$ _____, $C_{C1} =$ _____, $C_B =$ _____

Source	V_i	V_o	I_i	I_o	Z_{in}	Z_{out}	A_v	A_i	A_P	Power gain (dB)
Bead Board										

Review Questions:

- Since the current gain of CB amplifier is actually less than unity, what is the usefulness of this amplifier?

- Why two resistors are connected in parallel with base terminal, when emitter and collector require only one resistor?

- How a capacitor can be chosen from electrolytic and non electrolytic capacitor?

- Compare and contrast three configurations of the transistor?

Task:

Replicate the CB amplifier circuit in multisim by varying the values of R_1 , R_2 , R_E , R_C , C_{C1} one by one and explain the effect of changing the value of a given component respectively.

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 5

Name: _____ Roll No: _____

Score: _____ Signature of the Lab Tutor _____ Date: _____

LAB PERFORMANCE INDICATOR	Subject knowledge	Ability to conduct experiment	Observation/results	Presentation/ Report	Score
TOTAL	2	1	1	1	5
OBTAINED					

FET Small Signal Common Source Amplifier

Learning Outcomes:

At the conclusion of this experiment, understudy will be able to

- Determine the operation of a single stage Field-effect transistor when used as a common source (CS) small signal amplifier.
- Replicate the given CS circuit using kit and breadboard.
- Imitate the CS amplifier circuit design with a small change of parameters.

Required Equipment and Components:

Equipment	Components
Oscilloscope	Field Effect Transistor (FET)
General-purpose function generator	Few resistors
0 – ± 15 Power supply	Capacitors
Feed Back Trainer kit	
Breadboard	
Digital Multi-meter	

Theory:

The Field Effect Transistor (FET) is another significant kind of transistor other than the Bipolar Junction Transistor (BJT). FETs depend on an Electric Field to control the conductivity of a Channel in a semiconductor material. There are two general classes of FETs: the Metal-oxide-semiconductor FET (MOSFET) and the Junction FET (JFET).

The Junction Field Effect Transistor (JFET) can be designed as an Amplifier in which it works in linear amplification mode. Like the BJT, it can likewise be arranged as an Electronic Switch, in which it works on the other hand in cutoff and triode mode.

However, in numerous circuits FET is given benefit over BJT because of its capacity to give greater input impedance, voltage gain and more susceptible to temperature (which reduce noise level). FET can be utilized in phase-shift oscillators to minimize the loading effect.

There are three basic configurations of JFET Amplifier: the Common-Source (CS) Amplifier, the Common-Drain (CD) Amplifier and the Common-Gate (CG) Amplifier.

The CS Amplifier is the most versatile of the three basic JFET amplifier configurations due to its high input impedance and good voltage amplification. The common source mode of FET

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 5

connection is generally used audio frequency amplifiers and in high input impedance pre-amps and stages.

Mode of Operation:

For a Common-Source (CS) Amplifier, which is shown in Figure 1, the Input Signal is at the Gate, the Output Signal is at the Drain and the Source is at ground for AC signals due to the Bypass Capacitor CS. C1 is the Coupling Capacitor for the Input signal. The CS Amplifier is analogous to the BJT Common-Emitter Amplifier.

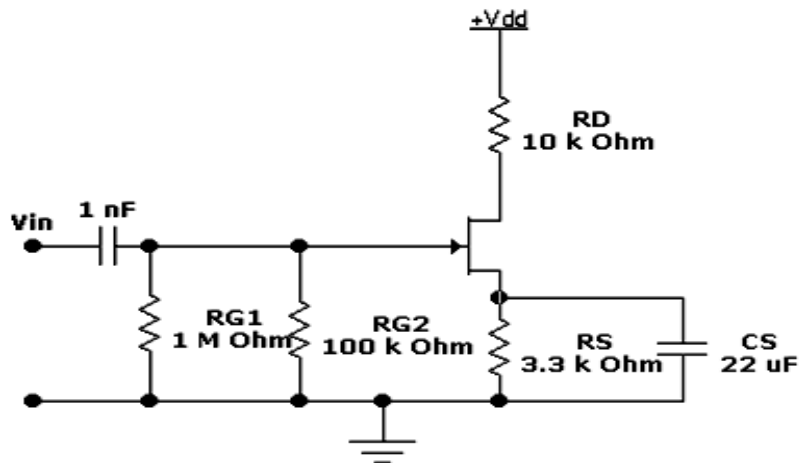


Figure 1: Circuit of a JFET Common-Source Amplifier

Characteristics of a CS Amplifier:

The CS Amplifier provides

- Almost an infinite Current Gain
- A moderate Voltage Gain (around -10).
- An extremely high Input Resistance (around 100 MΩ)
- A high Output Resistance (around 500 kΩ).
- A Phase Shift of 180° from Input to Output.

Procedure:

1. Utilizing the kit, make the connections according to Figure 1.
2. Give an input to the amplifier so that the output is 1 Vpp at 1 kHz, measure the input voltage (f=1 kHz).
3. Adjust the input signal through function generator until the output sine wave is no longer clipped which is being displayed through oscilloscope.
4. Observe the input and output voltage.
5. Then calculate the voltage gain of the amplifier.

Observation/Results:

CS configuration:

Transistor code: _____, Transistor type: _____ $R_G =$ _____, $R_D =$ _____ $R_S =$ _____, $C_{C1} =$ _____, $C_S =$ _____

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 5

Source	V_i	V_o	A_v
Kit			

Review Questions:

- Briefly explain how FETs utilize the voltage that is applied to their input terminal, called the Gate to control the current flowing through them resulting in the output current being proportional to the input voltage?

- Which type of biasing method is used in the above circuit?

- Explain in terms of mathematical equation, how FETs configured as CS amplifier have infinite current gain?

- What are the disadvantages of FET?

- What are the main constructional differences between a MOSFET and a BJT? What effect do they have on the current conduction mechanism of a MOSFET?

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 5

- What does it mean “the channel is pinched off”?

- A BJT is a current controlled device because its output characteristics are determined by the input current. A FET is voltage controlled device because its output characteristics are determined by the Field which depends on the Voltage applied. Since, current and voltage are generated in unison so why BJT is still called current controlled and FET is called voltage controlled?

Task:

Replicate the CS amplifier circuit in multisim by varying the values of R_{G1} , R_{G2} , R_D , R_S and C_{C1} one by one and explain the effect of changing the value of a given component respectively.

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 6

Name: _____ Roll No: _____

Score: _____ Signature of the Lab Tutor _____ Date: _____

LAB PERFORMANCE INDICATOR	Subject knowledge	Ability to conduct experiment	Observation/results	Presentation/ Report	Score
TOTAL	2	1	1	1	5
OBTAINED					

To observe the Frequency Response of Common Emitter Amplifier

Learning Outcomes:

At the conclusion of this experiment, understudy will be able to

- Introduce the concept of frequency response.
- Introduce the concept of DC and AC analysis.
- Introduce the high frequency model equivalent model for BJT.
- Determine the frequency response of BJT amplifier.

Required Equipment and Components:

Equipment	Components
Oscilloscope	Bipolar Junction Transistor (BJT), type 2N3904
General-purpose function generator	Few resistors
0 – +15 Power supply	Capacitors
Feed Back Trainer kit	
Breadboard	
Digital Multi-meter	

Theory:

Frequency Response:

Electrical engineers need to understand the intimate relationship between frequency and time. Circuits can be characterized both in time and frequency domain. It is important that an electrical engineer be able to travel between these two worlds effortlessly. Sometimes, a circuit problem that is difficult in one domain can be easily analyzed in the other domain. A good place to begin the study of the time-frequency relationship is in the design and analysis of a transistor amplifier circuit. An amplifier can be characterized by many properties. The most obvious one that comes to mind is gain. One would certainly like to know how much their circuit amplifies. However, gain is not constant with frequency. If it was, then one could use their stereo amplifier to amplify any electrical signal regardless of frequency, obviously this would reduce the demand for electronics engineers. Luckily, the relationship between gain and frequency can be quite complicated and ensures the continued employment of electronics engineers. Generally an amplifier has a lower limit and an upper limit to the frequencies that it can amplify. These limits can have many names such as lower -3dB point, upper -3dB point, F_l , F_h , lower corner frequency, upper corner frequency, etc. It is best to refer to these limits as the lower and upper -3dB frequencies or corner frequencies.

AC & DC Analysis:

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 6

a) DC Analysis:

As we know the frequency in DC is equal zero because there is no oscillation in DC signal so, the reactance of capacitor is equal infinity that mean the capacitors work as open circuit.

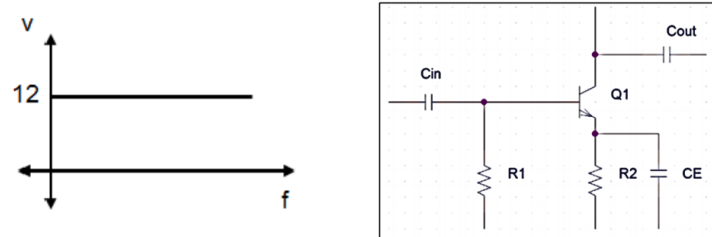


Figure 1: DC Analysis

b) AC Analysis:

In this analysis we will study the effect of capacitors in different level of frequencies that is at low, middle and high frequency.

Before we start our analysis, first we need to know the names of capacitors and its locations, there are two type of capacitor:

- Practical capacitors: Cin, Cout, CE these capacitors are connected as shown and it's effective in low frequency.
- Virtual capacitors: Cwi, Cwo, CBE, CBC, CCE these capacitors are appears and effective on the high frequency because its value in Nano-farad.

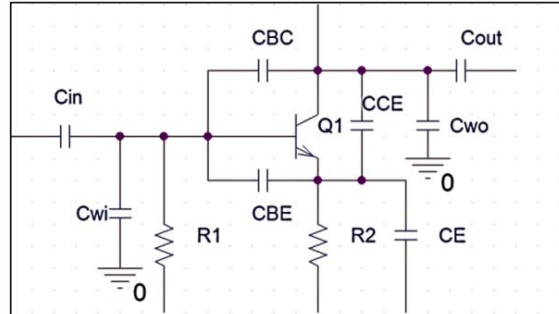


Figure 2: The sample circuit diagram of BJT amplifier

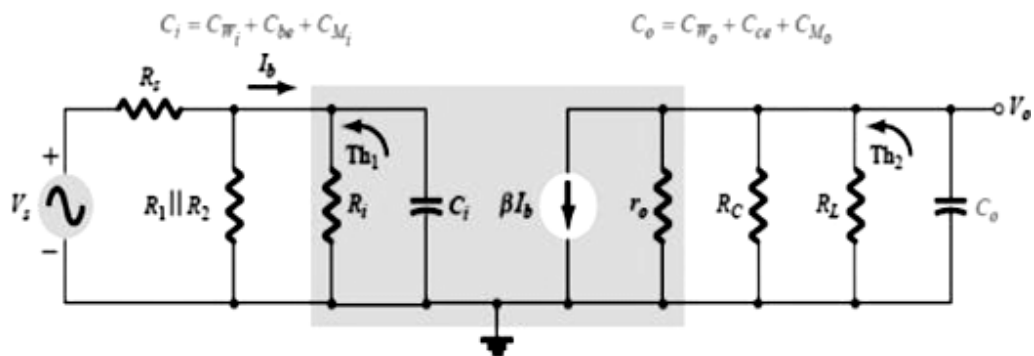


Figure 3: High frequency AC equivalent model

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 6

Cases of capacitor:

The relationship between frequency and capacitance is given by:

$$X_c = 1/2\pi fC$$

- At very low frequency the capacitor works as open circuit.
- At very high frequency the capacitor works as short circuit.
- At normal frequency the capacitor have a magnitude and effect on the frequency response.

Now, we will show the relationship between the voltage gain and varying frequency and defined the regions. The frequency response of BJT amplifier is shown in figure 4.

Frequency Response Curve of BJT amplifier:

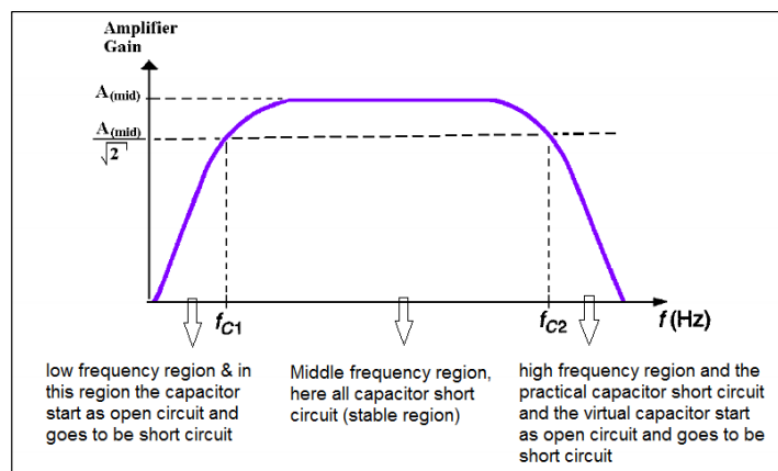


Figure 4: The frequency response of BJT amplifier

The frequency response begins with the lower frequency range designated between 0 Hz and lower cutoff frequency. At lower cutoff frequency, f_L , the gain is equal to $0.707 A_{mid}$. A_{mid} is a constant mid-band gain obtained from the mid-frequency range. The third, the higher frequency range covers frequency between upper cutoff frequency and above. Similarly, at higher cutoff frequency, f_H , the gain is equal to $0.707 A_{mid}$. Beyond this the gain decreases with frequency increases and dies off eventually.

The Lower Frequency Range: ($0 \rightarrow F_{low}$)

Since the impedance of coupling capacitors increases as frequency decreases, the voltage gain of a BJT amplifier decreases as frequency decreases. At very low frequencies, the capacitive reactance of the coupling capacitors may become large enough to drop some of the input voltage or output voltage. Also, the emitter-bypass capacitor may become large enough so that it no longer shorts the emitter resistor to ground.

The Higher Frequency Range: ($F_{high} \rightarrow \infty$)

The capacitive reactance of a capacitor decreases as frequency increases. This can lead to problems for amplifiers used for high-frequency amplification. The ultimate high cutoff frequency of an amplifier is determined by the physical capacitances associated with every component and of the physical wiring. Transistors have internal capacitances that shunt signal paths thus reducing the gain. The high cutoff frequency is related to a shunt time constant formed by resistances and capacitances associated with a node.

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 6

Procedure:

1. Utilizing the kit, make the connections as CE amplifier.
2. Set the amplitude of the input signal to by using the attenuation knob of the function generator.
3. Connect the oscilloscope at the output of the amplifier and observe the output waveform.
4. Now keep the amplitude constant and change the frequency to 1Hz.
5. Similarly change the frequencies logarithmically starting from 1Hz and up to 1MHz and fill the following table.

Observation/Results:

$$V_s = \text{_____} V$$

Frequency (Hz)	V_o (Volts)	Gain = V_o/V_s	Gain(dB) = $20 \log(V_o/V_s)$
10			
100			
1k			
10k			
50k			
100k			
1M			
2M			
5M			

Common Emitter Amplifier is studied and its Bandwidth is calculated.

1. Maximum Gain (A_{max}) = _____ dB
2. 3dB Gain = _____ dB
3. 3dB Lower cut-off frequency, f_L = _____ Hz
4. 3dB Upper cut-off frequency, f_H = _____ Hz
5. 3dB Bandwidth ($f_H - f_L$) = _____ Hz

Review Questions:

- What is Bandwidth of an amplifier?

- When modifying the given circuits, why one must only change the capacitance and not the resistance of the circuit?

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 6

- What do you mean by Miller theorem and Miller's effect?

- During the analysis of the CE amplifier's response over a wide range of frequency, why didn't you consider the Bypass capacitor?

- Explain using mathematical formula how capacitors behave as open circuit during DC analysis?

- Explain figure 4 in terms of mathematical formula that how capacitors behave open and short at low and high frequencies respectively?

Task:

Replicate the CC and CB amplifier circuit in multisim and explain the frequency response you observe by comparing it to CE amplifier's response over the range of the frequency you observed practically.

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 7

Name: _____ Roll No: _____

Score: _____ Signature of the Lab Tutor _____ Date: _____

LAB PERFORMANCE INDICATOR	Subject knowledge	Ability to conduct experiment	Observation/results	Presentation/ Report	Score
TOTAL	2	1	1	1	5
OBTAINED					

To observe the Frequency Response of Common Source Amplifier

Learning Outcomes:

At the conclusion of this experiment, understudy will be able to

- Determine the frequency response of FET amplifier.
- Determine the comparison between frequency responses of BJT and FET amplifier.

Required Equipment and Components:

Equipment	Components
Oscilloscope	Field Effect transistor (FET)
General-purpose function generator	Few resistors
0 – +15 Power supply	Capacitors
Feed Back Trainer kit	
Breadboard	
Digital Multi-meter	

Theory:

In general, BJT is much faster than the FET. This is the reason that BJT amplifiers can operate at wide range frequencies as compare to the FET amplifiers.

In an ideal amplifier, the gain is independent of frequency and the amplitude of the input signal. However from the practical point of view, the small signal amplifiers can provide a constant gain (that is still independent of the amplitude of the input signal) for a particular range of frequencies. The range at which they provide a constant gain is called pass-band. Below and after that range the gain decreases to a considerable amount. To determine the midband frequency region of the amplifier, the signals of same amplitude are applied at different frequencies. The experiments proved that the FET amplifier couldn't provide a constant gain at wide range of frequencies as the BJT amplifier did. Figure below shows the frequency response of FET amplifier.

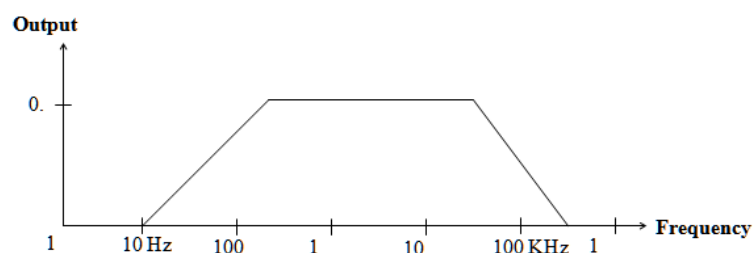


Figure 1: Frequency Response of FET Amplifier

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 7

Procedure:

1. Utilizing the kit, make the connections as CS amplifier.
2. Set the amplitude of the input signal to by using the attenuation knob of the function generator.
3. Connect the oscilloscope at the output of the amplifier and observe the output waveform.
4. Now keep the amplitude constant and change the frequency to 1Hz.
5. Similarly change the frequencies logarithmically starting from 10 Hz and up to 5 MHz and fill the following table.

Observation/Results:

$$V_s = \text{_____} V$$

Frequency(Hz)	V_o (Volts)	Gain = V_o/V_s	Gain(dB) = $20 \log(V_o/V_s)$
10			
100			
1k			
10k			
50k			
100k			
1M			
2M			
5M			

Common source amplifier is studied and its Bandwidth is calculated.

1. Maximum Gain (A_{max}) = _____ dB
2. 3dB Gain = _____ dB
3. 3dB Lower cut-off frequency, f_L = _____ Hz
4. 3dB Upper cut-off frequency, f_H = _____ Hz
5. 3dB Bandwidth ($f_H - f_L$) = _____ Hz

Review Questions:

- Why does FET make much less noise than BJT?

- Why BJT is faster than the FET in current mode signal processing?

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 7

- Why BJTs produce more gains comparative to FETs, when same amount of power has been utilized at the input?

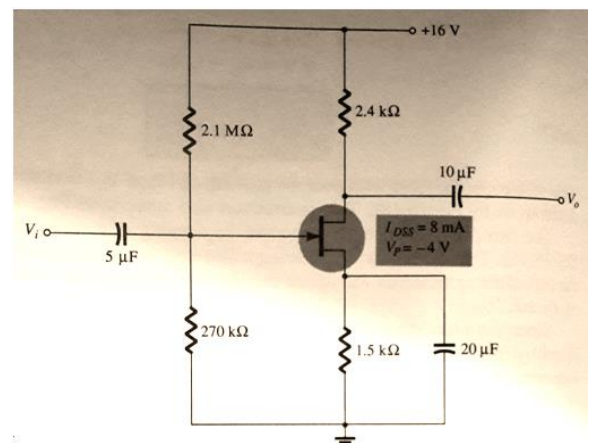
- Briefly explain what is meant by transconductance and voltage amplifier? Also, consider their characteristics.

- Common-source FET circuit which is analogous to BJT circuit may be viewed as a transconductance amplifier or as a voltage amplifier but among respective ideal conditions of these amplifiers what quality is not present in FET circuit?

- Why does FET have higher bandwidth than BJT?

Task:

Replicate the voltage divider biased common source amplifier circuit in multisim shown in figure. Omit the bypass capacitor and explain the frequency response you observe by comparing it to Self biased CS amplifier's response over the range of the frequency you observed practically.



Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 8

Name: _____ Roll No: _____

Score: _____ Signature of the Lab Tutor _____ Date: _____

LAB PERFORMANCE INDICATOR	Subject knowledge	Ability to conduct experiment	Observation/results	Presentation/ Report	Score
TOTAL	1	1.5	1.5	1	5
OBTAINED					

To observe the Operating Characteristics of a Push pull amplifier

Learning Outcomes:

At the conclusion of this experiment, understudy will be able to

- Introduce the concept of large signal amplifiers.
- Determine the operation of a push pull amplifier.
- Replicate the given push pull amplifier circuits using breadboard.

Required Equipment and Components:

Equipment	Components
Oscilloscope	Bipolar Junction Transistor (NPN and PNP)
General-purpose function generator	Fixed resistors
0 – ± 15 Power supply	Capacitors
Breadboard	
Digital Multi-meter	

Theory:

The main disadvantage of using small signal amplifier is their efficiency, as the majority of the power drawn from the supply is used by the amplifier itself, this goes against the primary purpose of a power amplifier, which is to deliver power drawn from supply to load. The solution comes in form of large signal amplifier.

Large Signal Amplifiers:

- Large Signal Amplifiers are also known as Power Amplifiers.
- Power Amplifiers can be sub-divided into different Classes, for example:
 - Class A Amplifiers – where the output device conducts for all of the input cycle.
 - Class B Amplifiers – where the output device conducts for only 50% of the input cycle.
 - Class AB Amplifiers – where the output device conducts for more than 50% but less than 100% of the input cycle.
- An ideal Power Amplifier would deliver 100% of the available DC power to the load.
- Class A amplifiers are the most common form of power amplifier but only have an efficiency rating of less than 40%.
- Class B amplifiers are more efficient than Class A amplifiers at around 70% but produce high amounts of distortion.
- Class B amplifiers consume very little power when there is no input signal present.
- By using the “Push-pull” output stage configuration, distortion can be greatly reduced.
- There are three classifications of Push-Pull Amplifier which are given below but generally Class B Amplifier is considered as Push Pull Amplifier.
 - Class A amplifier
 - Class B amplifier
 - Class AB amplifier
- However, simple push-pull Class B Power amplifiers can produce high levels of Crossover Distortion due to their cut-off point biasing.

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 8

- A common method to remove that crossover distortion in Class B amplifier is to bias both the transistor at a point slightly above then the cut-off point of transistor. Then this circuit is known as Class AB amplifier circuit

Push Pull Amplifier:

Push-pull amplifiers use two “complementary” or matching transistors, one being an NPN-type and the other being a PNP-type with both power transistors receiving the same input signal together that is equal in magnitude, but in opposite phase to each other. This results in one transistor only amplifying one half or 180° of the input waveform cycle while the other transistor amplifies the other half or remaining 180° of the input waveform cycle with the resulting “two-halves” being put back together again at the output terminal. Then the conduction angle for this type of amplifier circuit is only 180° or 50% of the input signal. This pushing and pulling effect of the alternating half cycles by the transistors gives this type of circuit its amusing “push-pull” name.

The Class AB amplifier circuit is the combination of both Class A and Class B amplifier. By adding the diode, transistors are biased in slightly conducting state even when no signal is present at base terminal, thus removing the crossover distortion problem. A small collector current will flow when there is no input signal but it is much less than that for the Class A amplifier configuration. This means then that the transistor will be “ON” for more than half a cycle of the waveform but much less than a full cycle giving a conduction angle of between 180° to 360° or 50% to 100% of the input signal depending upon the amount of additional biasing used.

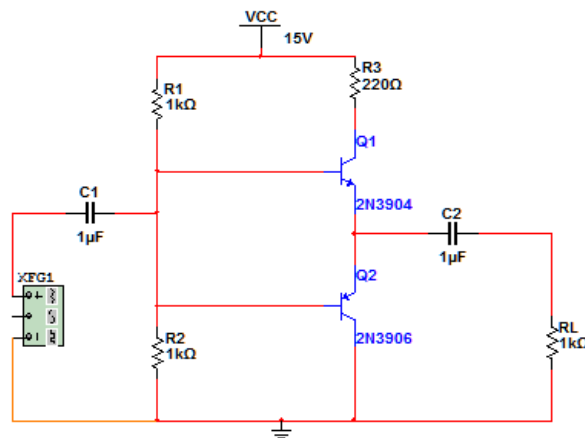


Figure 1: Class B Push-pull amplifier

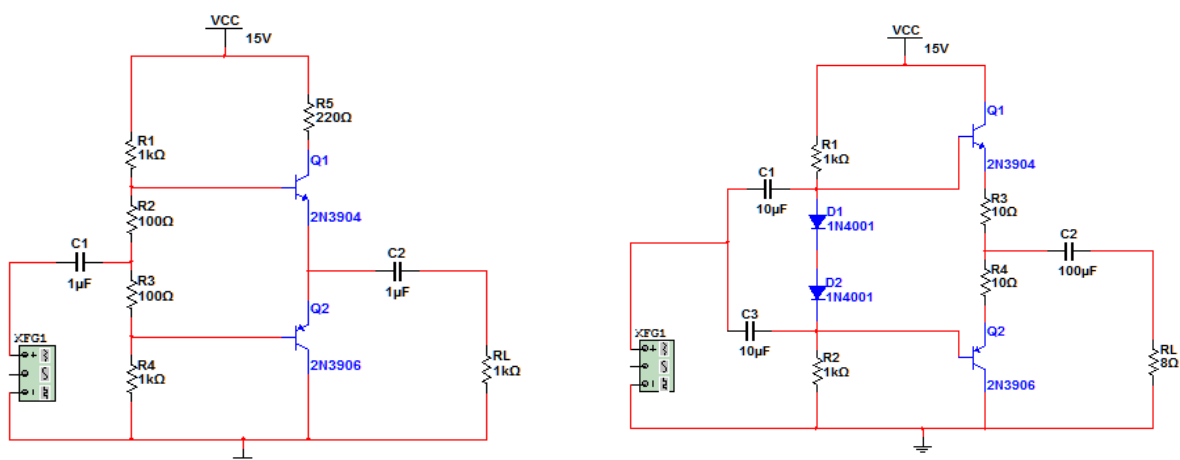


Figure 2: Class AB Push-pull amplifier with resistor and diode biasing

Procedure:

1. Make the circuit as shown in figure 1.
2. Record the output voltage without the input signal.
3. Then record the output voltage by applying the input signal. Observe the distortion in output waveform
4. Now, make the circuit as shown in figure 2.
5. Replicate the steps 2-3; also observe the distortion less output waveform.

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 8

Observation/Results:

Vi	Class B (Vo)	Class AB (Vo)
Without Signal		
With Signal		

Review Questions:

- What is the main source of distortion in a push pull amplifier?
- Briefly explain why do you think Q1 is in phase and Q2 is out of phase?
- Write the advantages and disadvantages of Push Pull Amplifier?

Task:

- Calculate the Q-Points of Class B and AB Push-pull amplifier circuit shown in figure 1 and 2 respectively.
- Replicate the Class B and AB Push-pull amplifier circuit in multisim shown in figure 1 and 2 respectively. Replace the ordinary NPN/PNP transistors by power transistors BD329/BD330 respectively. Set the input voltage as 1.5Vp. Compare the output waveforms at different frequencies to observe distortion in class B push pull amplifier while class AB has distortion less output waveform. Also, use measurement probe across input and RL to observe ac and dc values of current as well as voltage. Attach the screenshot of waveforms.
- Imitate the Push-pull amplifier circuit utilizing the Darlington and feedback pair in multisim as shown below in figure 3 and 4 respectively in order to observe their characteristics.

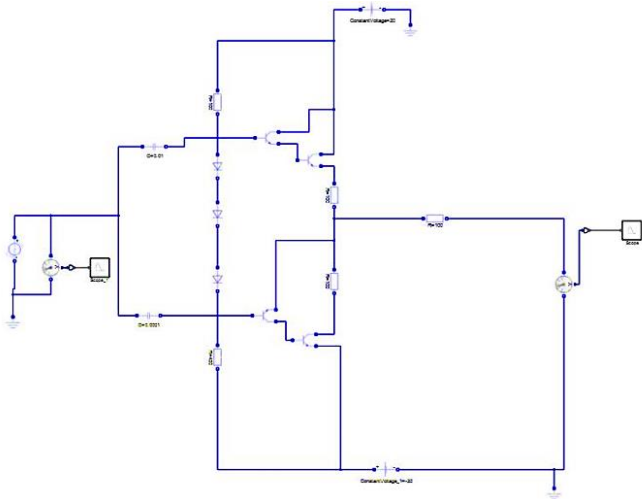


Figure 3: Push-pull amplifier using Darlington pair

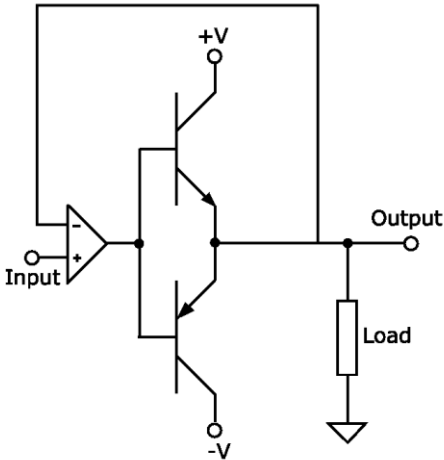


Figure 4: Push-pull amplifier using feedback pair

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 9

Name: _____ Roll No: _____

Score: _____ Signature of the Lab Tutor _____ Date: _____

LAB PERFORMANCE INDICATOR	Subject knowledge	Ability to conduct experiment	Observation/results	Presentation/ Report	Score
TOTAL	1	1	2	1	5
OBTAINED					

To observe the Operating Characteristics of a Tuned Amplifier

Learning Outcomes:

At the conclusion of this experiment, understudy will be able to

- Introduce the concept of Tuning.
- Determine the operation of a Tuned amplifier.
- Replicate the given Tuned amplifier circuit using kit.

Required Equipment and Components:

Equipment	Components
Oscilloscope	Variable Inductor/coil
General-purpose function generator	Variable Capacitor
Feed Back Trainer kit	
Power supply	

Theory:

The sorts of amplifiers that we have talked about so far can't work successfully at radio frequencies, despite the fact that they are great at audio frequencies. Also, the gain of these amplifiers is such that it will not vary according to the frequency of the signal, over a wide range. This allows the amplification of the signal equally well over a range of frequencies and does not permit the selection of particular desired frequency while rejecting the other frequencies. So, there occurs a need for a circuit which can select as well as amplify. So, an amplifier circuit along with a selection, such as a tuned circuit makes a Tuned amplifier.

Tuned Amplifier:

Tuned amplifiers are the amplifiers that are employed for the purpose of tuning. Tuning means selecting. Among a set of frequencies available, if there occurs a need to select a particular frequency, while rejecting all other frequencies, such a process is called Selection. This selection is done by using a circuit called as Tuned circuit. The tuned amplifier circuit is just like BJT/FET voltage-divider circuit with an exception that a TANK circuit replaces the R_c/R_d . A TANK circuit is the parallel combination of an inductor and a capacitor, as shown in figure 1.

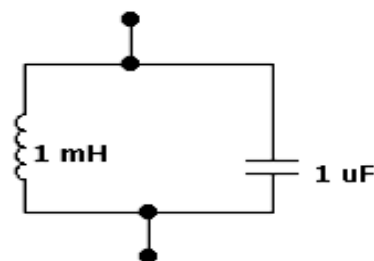


Figure 1: L.C Tank

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 9

When an amplifier circuit has its load replaced by a tuned circuit, such an amplifier can be called as a Tuned amplifier circuit. The basic tuned amplifier circuit looks as shown in figure 2.

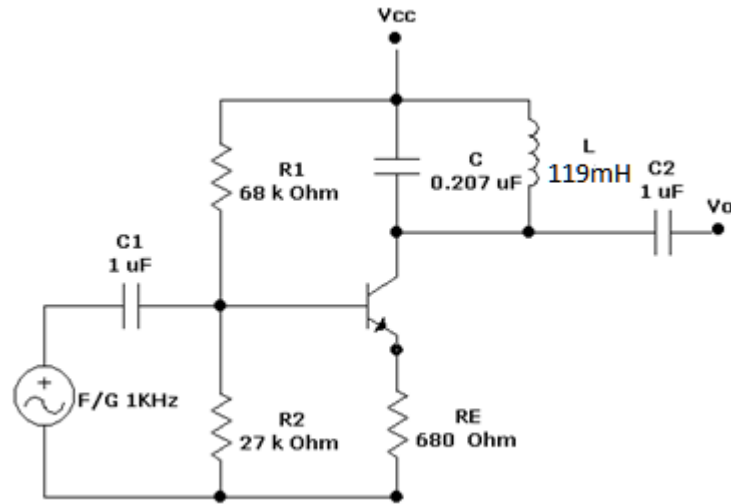


Figure 2: Simple Tuned Amplifier

Tuned amplifier provide high constant gain over a specified bandwidth centered at a specified frequency f and as much rejection as possible outside this band practical gain and phase characteristics are shown in Figure 3.

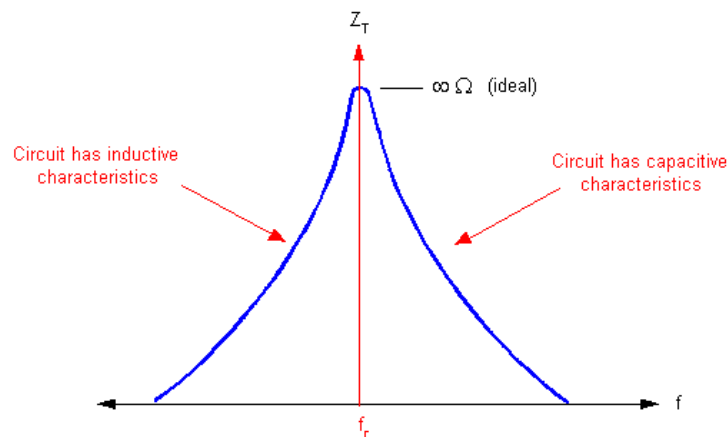


Figure 3: Frequency Response of Tuned Amplifier

If the frequency increases then X_C (capacitance reactance) decreases and X_L (inductance reactance) increases. When the reactance of the inductor balances the reactance of the capacitor, in the tuned circuit at some frequency, such a frequency can be called as resonant frequency. It is denoted by f_r . At frequencies below resonance, the tank circuit is inductive and above it, is capacitive. The formula for resonance is

$$2\pi f_L = \frac{1}{2\pi f_c} \quad \Rightarrow \quad f_r = \frac{1}{2\pi\sqrt{LC}}$$

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 9

The very important factor which is responsible for affecting the bandwidth of a Tuned amplifier is “Q-factor” or “Quality-factor of the tuned amplifier”. It could be defined as the “ratio of center frequency (f_o) to the bandwidth”.

i.e. $Q = \frac{f_o}{BW}$ Or $Q = \frac{1}{2\pi f_o RC}$ (It is unit less measure of the quality of the coil).

Where $BW = f_2 - f_1$

Tuned circuit can be used as a load for class B and class C power amplifier operation. It eliminates the harmonics, and then the output will be a pure sine wave although the collector current is distorted. A single stage tuned amplifier is seldom used in practice. Generally, a double-tuned type is preferred, due to using it in cascaded amplification.

Procedure:

1. Connect the circuit as shown in Figure 2.
2. Set the input signal V_i by using the Signal Generator.
3. Note down the values of L and C where resonant frequency occurs practically.
4. Now, keep the input voltage constant and note down the practical values of frequency by varying one parameter at a time that is, L, and C respectively.
5. Also, note down the values of voltage at respective values of L and C.

Observation/Results:

$V_i = \underline{\hspace{2cm}} V$

L	C	Theoretical frequency f_o (KHz)	Practical frequency f_o (KHz)	V_o

Review Questions:

- While performing the practical at which values of L and C, you observed the resonant frequency? How did you decide that it is the resonant frequency for the respective amplifier?

- How does tank circuit behaves as an open-circuit at resonant frequency?

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 9

-
-
- What are the factors that can affect the performance of the Tuned amplifier?
-
-
-

- Why tuned amplifier cannot be used at low frequency?
-
-
-

- What is the other name used for tuned amplifier?
-
-
-

- Define Single tuned and double tuned amplifier?
-
-
-

- What is Stagger tuned amplifier?
-
-
-

Task:

Replicate the Single Tuned amplifier circuit in multisim. By taking 3 different values of L and C Calculate f_0 , Q, BW and gain at resonant theoretically and compare your results with software results obtained. Then, briefly explain your observations in terms of frequency response.

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 10

Name: _____ Roll No: _____

Score: _____ Signature of the Lab Tutor _____ Date: _____

LAB PERFORMANCE INDICATOR	Subject knowledge	Ability to conduct experiment	Observation/results	Presentation/ Report	Score
TOTAL	1	2	1	1	5
OBTAINED					

To Understand the Operation of R-C Coupled Two Stage Bipolar Junction Transistor Amplifiers

Learning Outcomes:

At the conclusion of this experiment, understudy will be able to

- Introduce the concept of multistage amplifiers.
- Determine the multistage amplifiers on the basis of coupling.
- Replicate the given R-C coupled two-stage BJT amplifier circuit using breadboard.

Required Equipment and Components:

Equipment	Components
Oscilloscope	Bipolar Junction Transistor
General-purpose function generator	Few Resistors
0 – ±15 Power supply	Few Capacitors
	Connecting Leads

Theory:

Multi-stage transistor amplifiers:

In practical applications, the output of a single state amplifier is usually insufficient, though it is a voltage or power amplifier. Hence they are replaced by Multi-stage transistor amplifiers.

In Multi-stage amplifiers, the output of first stage is coupled to the input of next stage using a coupling device. These coupling devices can usually be a capacitor or a transformer. This process of joining two amplifier stages using a coupling device can be called as Cascading. The following figure shows a two-stage amplifier connected in cascade.



The overall gain is the product of voltage gain of individual stages.

$$A_V = A_{V1} \times A_{V2} = \frac{V_2}{V_1} \times \frac{V_0}{V_2} = \frac{V_0}{V_1}$$

Where A_V = Overall gain, A_{V1} = Voltage gain of 1st stage, and A_{V2} = Voltage gain of 2nd stage.

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 10

If there is **n** number of stages, the product of voltage gains of those **n** stages will be the overall gain of that multistage amplifier circuit.

The following parameters should be maintained properly during the process of cascading:

- **Impedance Matching:** When coupling amplifier stages together impedance matching is important so that as much signal as possible is transferred from the output of one stage to the input of the next, keeping inter stage losses to a minimum.
- **Correct Frequency Response:** Ensuring that the correct bandwidth is maintained throughout all stages of amplification.
- **DC Isolation:** Whether it is required or not. It may be required that where the output of one stage is at a different DC potential to the input of the next, the two stages are electrically isolated from each other.

Classification of amplifiers in terms of coupling:

Since, joining one amplifier stage with the other in cascade, using coupling devices form a Multi-stage amplifier circuit. Following are the basic methods of coupling,

- Resistance-Capacitance Coupling
- Impedance Coupling
- Transformer Coupling
- Direct Coupling

Resistance-capacitance (RC) coupling is most widely used to connect the output of first stage to the input (base) of the second stage and so on. It is the most popular type of coupling because it is cheap and provides a constant amplification over a wide range of frequencies. These R-C coupled amplifier circuits are commonly used as voltage amplifiers in the audio systems. Figure 1 shows the circuit diagram of the R-C coupled two-stage BJT amplifier.

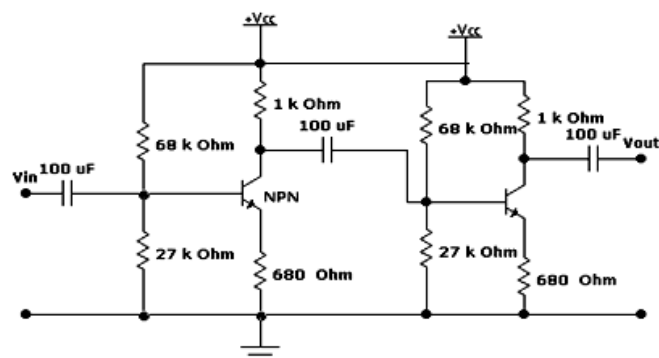


Figure 1: R-C coupled two-stage BJT amplifier

Procedure:

1. Make a circuit as shown in figure 1.
2. Set the input signal V_i by using the Signal Generator.
3. Initially calculate the gain of the first stage, and then calculate the gain of the second stage.
4. Get the product of the two stages.
5. Take at least 3 readings by varying input signal.

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 10

Observation/Results:

Input Signal	Gain (A_{v1}) 1 st stage	Gain (A_{v2}) 2 nd stage	Overall Gain $A_v = A_{v1} \times A_{v2}$	A_v Measured

Review Questions:

- Briefly explain the difference between following Classifications of amplifiers on the basis of coupling: Resistance-Capacitance, Impedance, Transformer and Direct Coupling?

- What are the two basic purposes of a coupling device?

- What is the effect on the total bandwidth, if amplifiers are connected in cascaded form?

Task:

Replicate the two-stage CS JFET amplifier circuit in multisim shown in figure and explain the gain you observe at both stages by comparing it to the two-stage CE BJT amplifier's gain over the range of the input signal you observed practically.

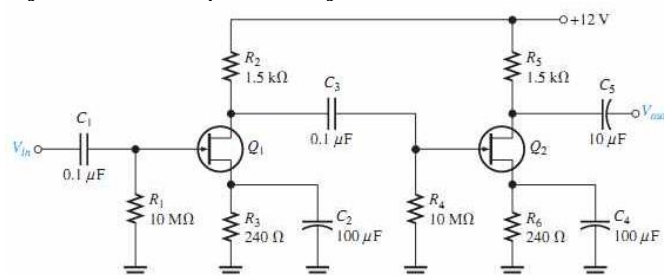


Figure 2: A two-stage CS JFET amplifier circuit.

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 11

Name: _____ Roll No: _____

Score: _____ Signature of the Lab Tutor _____ Date: _____

LAB PERFORMANCE INDICATOR	Subject knowledge	Ability to conduct experiment	Observation/results	Presentation/ Report	Score
TOTAL	1	2	1	1	5
OBTAINED					

To Understand the Operation of Colpitts Oscillator

Learning Outcomes:

At the conclusion of this experiment, understudy will be able to

- Introduce the concept of an electronic oscillator and its principal types.
- Introduce the concept of LC electronic oscillators.
- Determine the operation of Colpitts Oscillator.
- Replicate the given oscillator circuit using kit.

Required Equipment and Components:

Equipment	Components
Oscilloscope	Connecting Leads
F.B Trainer kit	Probes
L.C tank	
0 – ± 15 Power supply	

Theory:

Oscillator:

- An oscillator is an electronic circuit that produces a periodic, oscillating electronic signal, often a sine wave or a square wave. Oscillators convert direct current (DC) from a power supply to an alternating current (AC) signal.
- The main requirement of an oscillator is that the output should be uniform and may not vary in frequency or amplitude.
- A tank circuit is formed when a capacitor is connected in parallel with an inductor and oscillates when an external voltage source is applied. The oscillations of a tank circuit are dampened by the resistance of the circuit. For a tank circuit to maintain oscillation, positive feedback is required.
- An oscillator has three basic parts: a frequency- determining device, an amplifier, and a feedback circuit.
- There are two main types of electronic oscillator – the linear or harmonic oscillator and the nonlinear or relaxation oscillator.
- The harmonic, or linear, oscillator produces a sinusoidal output. There are two types: Feedback oscillator and Negative-resistance oscillator.
- Feedback oscillators are further classified as LC oscillators, crystal oscillators, and RC oscillators. Typical LC oscillator circuits are the Hartley, Colpitts and Clapp circuits.

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 11

Although Crystal oscillators provide more stability than LC oscillators while RC oscillators use resistance-capacitance networks to determine the oscillator frequency.

- A nonlinear or relaxation oscillator produces a non-sinusoidal output, such as a square, saw tooth or triangle wave. A relaxation oscillator stores energy in a reactive component during one part of the oscillation cycle. Examples of relaxation oscillators are blocking oscillators and multi-vibrators.
- Oscillators are widely used in many electronic devices ranging from simplest clock generators to digital instruments (like calculators) and complex computers and peripherals etc. Common examples of signals generated by oscillators include signals broadcast by radio and television transmitters, clock signals that regulate computers and quartz clocks, and the sounds produced by electronic beepers and video games.

Colpitts Oscillator:

- The Colpitts circuit, like other LC oscillators, consists of a gain device (such as a bipolar junction transistor, field-effect transistor, operational amplifier, or vacuum tube) with its output connected to its input in a feedback loop containing a parallel LC circuit (tuned circuit), which functions as a band pass filter to set the frequency of oscillation.
- The emitter terminal of the transistor is effectively connected to the junction of the two capacitors, C1 and C2 which are connected in series and act as a simple voltage divider. When the power supply is firstly applied, capacitors C1 and C2 charge up and then discharge through the coil L. The oscillations across the capacitors are applied to the base-emitter junction and appear in the amplified at the collector output.
- Resistors, R1 and R2 provide the usual stabilizing DC bias for the transistor in the normal manner while the additional capacitors act as DC-blocking bypass capacitors.
- The required external phase shift is obtained in a way that the configuration of the transistor amplifier is of a Common Emitter Amplifier with the output signal 180° out of phase with regards to the input signal. The additional 180° phase shift require for oscillation is achieved by the fact that the two capacitors are connected together in series but in parallel with the inductive coil resulting in overall phase shift of the circuit being zero or 360°.
- The amount of feedback is determined by the ratio of C1 and C2. These two capacitances are generally “ganged” together to provide a constant amount of feedback so that as one is adjusted the other automatically follows.
- This oscillator is designed for the generation of high frequency sinusoidal oscillations with the radio frequencies typically ranging from 10 KHz to 300MHz.
- The frequency of oscillations for a Colpitts oscillator is determined by the resonant frequency of the LC tank circuit and is given as:

$$f_0 = \frac{1}{2\pi\sqrt{L\frac{C_1C_2}{C_1+C_2}}}$$

$$f_r = \frac{1}{2\pi\sqrt{LC_T}}$$

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 11

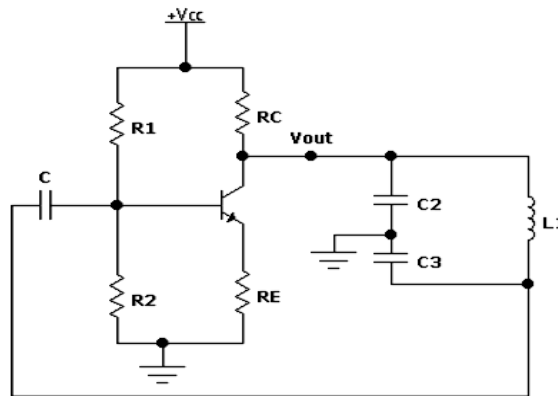


Figure 1: Colpitts Oscillator

Procedure:

1. Make a circuit as shown in figure 1. Select proper values of L1, C2 and C3.
2. Connect CRO at output terminals and observe the wave form.
3. Calculate practically the frequency of oscillations by using the expression.

$$f = \frac{1}{T}$$
, where T= number of space $\times$ time/div
4. Repeat the above steps 2, 3 by varying one parameter at a time that is, L1, C2 and C3 respectively and note down the practical values of oscillations of Colpitts oscillator.
5. Compare the values of frequency of oscillations both theoretically and practically.
6. Also, calculate the gain (A).

Observation/Results:

L1	C2	C3	Theoretical frequency (KHz)	Practical frequency (KHz)	AV

Review Questions:

- What is the general condition of oscillation?

- What is the loop gain of an oscillator?

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 11

- What is the difference between an amplifier and an oscillator?

- What are the applications of a Colpitts Oscillator?

- What is the difference between Saw tooth and Triangular wave?

- What is the advantage of using this type of capacitive circuit configuration?

Task:

Imitate the Colpitts Oscillator by using an op-amp in multisim as shown below in figure 2. Observe its principal characteristics by adjusting the gain of the op-amp using the feedback resistors R_i and R_f that is, $A_v = -R_f / R_i$

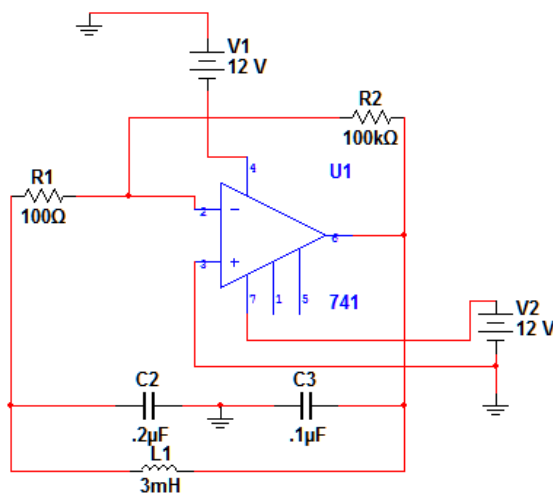


Figure 2: Colpitts Oscillator by using op-amp

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 12

Name: _____ Roll No: _____

Score: _____ Signature of the Lab Tutor _____ Date: _____

LAB PERFORMANCE INDICATOR	Subject knowledge	Ability to conduct experiment	Observation/results	Presentation/ Report	Score
TOTAL	2	1	1	1	5
OBTAINED					

To Understand the Operation of Hartley Oscillator

Learning Outcomes:

At the conclusion of this experiment, understudy will be able to

- Determine the operation of Hartley Oscillator.
- Determine the comparison between Hartley and Colpitts Oscillator.
- Replicate the given oscillator circuit using kit.

Required Equipment and Components:

Equipment	Components
Oscilloscope	Connecting Leads
F.B Trainer kit	Probes
L.C tank	
0 – +15 Power supply	

Theory:

Hartley oscillator is very popular and is commonly used as a local oscillator in radio receivers. The Hartley oscillator is an LC electronic oscillator that derives its feedback from a tapped coil in parallel with a capacitor (the tank circuit). The tank circuit is made up of L1, L2, and C1. The coil L1 is inductively coupled to coil L2, the combination of two functions acts as auto transformer. The resistances R1 and R2 provide the necessary biasing. The capacitance C2 and C3 blocks the d.c component. The frequency of oscillations is determined by the values of L1, L2 and C1 and is given by,

$$f_o = \frac{1}{2\pi\sqrt{C1(L1 + L2)}}$$

Positive feedback can be earned by the mutual inductance between L1 of transistor output circuit and L2 of base emitter input circuit. In CE circuit, the phase inversion of 180° occurs always between the collector and the base voltage. Also, with the inductor tap connected to the emitter (common terminal of transistor) as the base (L1-L2 combination), another phase inversion of 180° occurs at the opposite terminals of the inductor. Therefore, if these two are combined, the feedback condition of same phase is realized, and the closed circuit gain is earned by Q1.

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 12

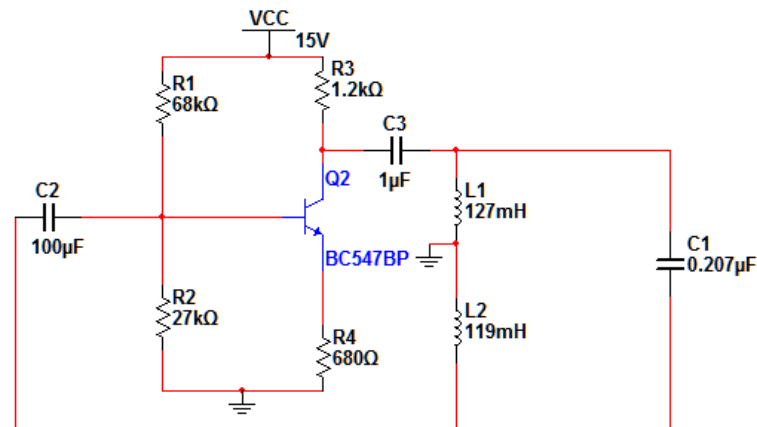


Figure 1: Hartley Oscillator

Advantages of the Hartley oscillator:

- The frequency may be adjusted using a single variable capacitor, one side of which can be earthed.
- The output amplitude remains constant over the frequency range.
- Either a tapped coil or two fixed inductors are needed, and very few other components.
- Easy to create an accurate fixed-frequency crystal oscillator variation by replacing the capacitor with a (parallel-resonant) quartz crystal or replacing the top half of the tank circuit with a crystal and grid-leak resistor (as in the Tri-tet oscillator).

Disadvantages of the Hartley oscillator:

Harmonic-rich output if taken from the amplifier and not directly from the LC circuit (unless amplitude-stabilization circuitry is employed).

Procedure:

1. Make a circuit as shown in figure 1. Select proper values of C1, L1 and L2.
2. Connect CRO at output terminals and observe the wave form.
3. Calculate practically the frequency of oscillations by using the expression.

$$f = \frac{1}{T}, \text{ where } T = \text{number of space} \times \text{time/div}$$
4. Repeat the above steps 2, 3 by varying one parameter at a time that is, C1, L1 and L2 respectively and note down the practical values of oscillations of Hartley oscillator.
5. Compare the values of frequency of oscillations both theoretically and practically.
6. Also, calculate the gain (A).

Observation/Results:

C1	L1	L2	Theoretical frequency (KHz)	Practical frequency (KHz)	AV

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 12

Review Questions:

- One of the main disadvantages of the basic LC Oscillator circuit is that they have no means of controlling the amplitude of the oscillations and also, it is difficult to tune the oscillator to the required frequency. Consider the cases what happens if the cumulative electromagnetic coupling between L_1 and L_2 is too small? Likewise, what happens if the feedback is too strong?

- Explain the process of Automatic Base Bias?

- Which one is preferred more Hartley or Colpitts Oscillator? and why?

Task:

Imitate the Hartley Oscillator by using an op-amp in multisim as shown below in figure 2. Observe its principal characteristics by adjusting the gain of the op-amp using the feedback resistors R_i and R_f that is, $A_v = -R_f / R_i$

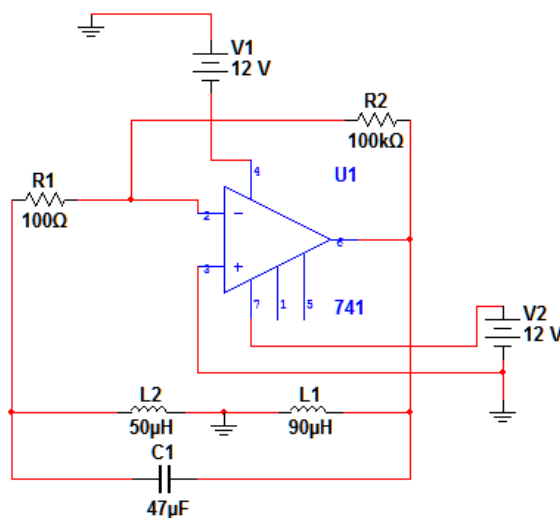


Figure 2: Hartley Oscillator by using op-amp

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 13

Name: _____ Roll No: _____

Score: _____ Signature of the Lab Tutor _____ Date: _____

LAB PERFORMANCE INDICATOR	Subject knowledge	Ability to conduct experiment	Observation/results	Presentation/ Report	Score
TOTAL	2	1	1	1	5
OBTAINED					

To Understand the Operation of RC-Phase Shift Oscillator

Learning Outcomes:

At the conclusion of this experiment, understudy will be able to

- Introduce the concept of RC electronic oscillators.
- Determine the operation of Resistance-Capacitance oscillator.
- Replicate the given oscillator circuit using breadboard.

Required Equipment and Components:

Equipment	Components
Oscilloscope	Op-Amp IC
Bread Board	Few Resistors
0 – ± 15 Power supply	Few Capacitors
	Connecting Leads

Theory:

As a single stage transistor amplifier can produce 180° of phase shift between its output and input signals when connected in a class-A type configuration.

For an oscillator to sustain oscillations indefinitely, sufficient feedback of the correct phase, that is “Positive Feedback” must be provided along with the transistor amplifier being used acting as an inverting stage to achieve this.

- In a Resistance-Capacitance Oscillator or simply an RC Oscillator circuit the input is shifted 180° through the amplifier stage and 180° again through a second inverting stage giving us “ $180^\circ + 180^\circ = 360^\circ$ ” of phase shift which is effectively the same as 0° thereby giving us the required positive feedback. In other words, the phase shift of the feedback loop should be “0”.
- The basic RC Oscillator which is also known as a Phase-shift Oscillator, produces a sine wave output signal using regenerative feedback obtained from the resistor-capacitor combination. This regenerative feedback from the RC network is due to the ability of the capacitor to store an electric charge, (similar to the LC tank circuit).
- This resistor-capacitor feedback network can be connected to produce a leading phase shift (phase advance network) or interchanged to produce a lagging phase shift (phase retard network) as shown below, the outcome is still the same as the sine wave oscillations only occur at the frequency at which the overall phase-shift is 360° .

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 13

- By varying one or more of the resistors or capacitors in the phase-shift network, the frequency can be varied and generally this is done by keeping the resistors the same and using a 3-ganged variable capacitor.
- If all the resistors, R and the capacitors, C in the phase shift network are equal in value, then the frequency of oscillations produced by the RC oscillator is given as:

$$f_r = \frac{1}{2\pi RC\sqrt{2N}}$$

Where:

f_r is the Output Frequency in Hertz

R is the Resistance in Ohms

C is the Capacitance in Farads

N is the number of RC stages. (N = 3)

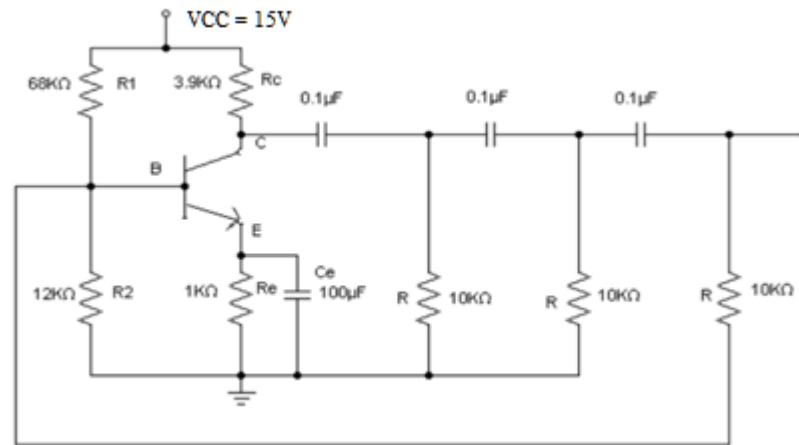


Figure 1: RC-Phase Shift Oscillator with phase retard network

Since the resistor-capacitor combination in the RC Oscillator circuit also acts as an attenuator producing a total attenuation of $-1/29$ th ($V_o/V_i = \beta$) across the three stages, the voltage gain of the amplifier must be sufficiently high enough to overcome these RC losses.

The loading effect of the amplifier on the feedback network has an effect on the frequency of oscillations and can cause the oscillator frequency to be up to 25% higher than calculated. Then the feedback network should be driven from a high impedance output source and fed into a low impedance load such as a common emitter transistor amplifier but better still is to use an Operational Amplifier as it satisfies these conditions perfectly.

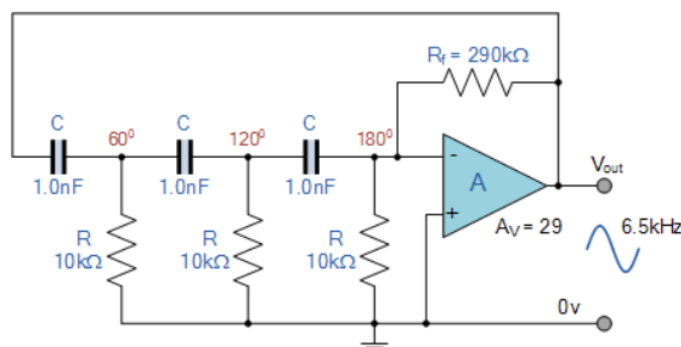


Figure 2: RC-Phase Shift Oscillator by using op-amp

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 13

Procedure:

1. Make a circuit as shown in figure 2.
2. Connect CRO at output terminals and observe the wave form.
3. Calculate practically the frequency of oscillations by using the expression.
$$f = \frac{1}{T}, \text{ where } T = \text{number of space} \times \text{time/div}$$
4. Repeat the above steps 2, 3 by varying one parameter at a time that is, R, and C respectively and note down the practical values of oscillations of RC-Phase Shift.
5. Compare the values of frequency of oscillations both theoretically and practically.
6. Also, calculate the gain (A).

Observation/Results:

R	C	Theoretical frequency (KHz)	Practical frequency (KHz)	AV

Review Questions:

- What is the difference between damped oscillations and undamped oscillations?

- What are the drawbacks of RC phase shift oscillator?

- Why three stages of RC network have been used in a way that the output voltage leads the input voltage by an angle of about 60° as an ideal single-pole RC circuit would produce a phase shift of exactly 90°. So what happens if two single-poles used in an RC oscillator in order to give required 180° phase shift?

Task:

Imitate the RC-Phase Shift Oscillator by using transistor in multisim as shown below in figure 1. Observe its principal characteristics.

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 14

Name: _____ Roll No: _____

Score: _____ Signature of the Lab Tutor _____ Date: _____

LAB PERFORMANCE INDICATOR	Subject knowledge	Ability to conduct experiment	Observation/results	Presentation/ Report	Score
TOTAL	2	1	1	1	5
OBTAINED					

To understand the operation of Wien-Bridge oscillators

Learning Outcomes:

At the conclusion of this experiment, understudy will be able to

- Determine the operation of Wien-Bridge oscillators.
- Determine the comparison between RC Phase-shift and Wien-Bridge Oscillator.
- Replicate the given oscillator circuit using breadboard.

Required Equipment and Components:

Equipment	Components
Oscilloscope	Op-Amp IC
Bread Board	Few Resistors
0 – ± 15 Power supply	Few Capacitors
	Connecting Leads

Theory:

- The Wien Bridge Oscillator uses a RC network in place of the conventional LC tuned tank circuit to produce a sinusoidal output waveform and is so called because the circuit is based on a frequency-selective form of the Wheatstone bridge circuit.
- The Wien Bridge oscillator is a two-stage RC coupled amplifier circuit that has good stability at its resonant frequency, low distortion and is very easy to tune making it a popular circuit as an audio frequency oscillator but the phase shift of the output signal is considerably different from the phase shift RC Oscillator.
- The Wien Bridge Oscillator uses a feedback circuit consisting of a series RC circuit (High Pass Filter) connected with a parallel RC (Low Pass Filter) of the same component values giving a very selective second-order frequency dependant Band Pass Filter with a high Q factor at that selected frequency, f_r . At the resonant frequency f_r the phase shift is 0° . Consider the circuit below.
- At low frequencies the reactance of the series capacitor (C_1) is very high so acts a bit like an open circuit, blocking any input signal at V_{in} resulting in virtually no output signal, V_{out} . Likewise, at high frequencies, the reactance of the parallel capacitor, (C_2) becomes very low, so this parallel connected capacitor acts a bit like a short circuit across the output, so again there is no output signal.
- So there must be a frequency point between these two extremes of C_1 being open-circuited and C_2 being short-circuited where the output voltage, V_{OUT} reaches its maximum value.

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 14

The frequency value of the input waveform at which this happens is called the oscillators Resonant Frequency, (f_r).

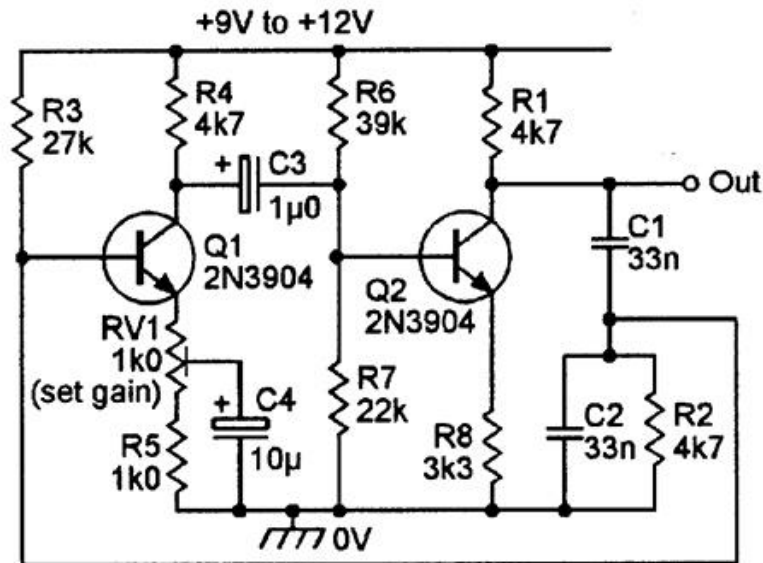


Figure 1: Wien-Bridge Oscillator

- At this resonant frequency, the circuit's reactance equals its resistance, that is: $X_c = R$, and the phase difference between the input and output equals zero degrees. The magnitude of the output voltage is therefore at its maximum and is equal to one third ($1/3$) of the input voltage as shown.

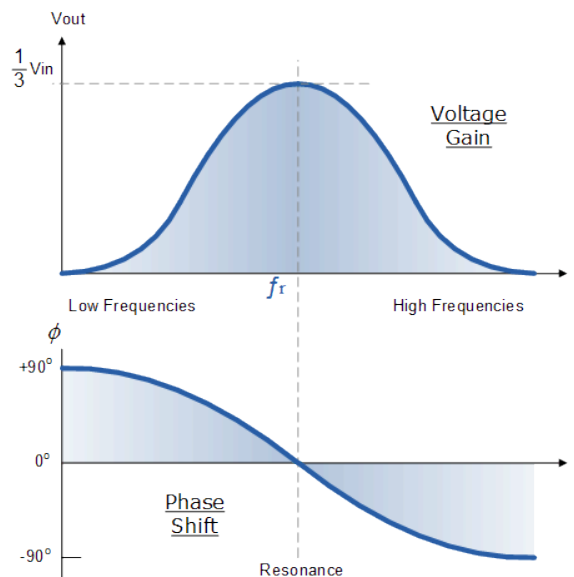


Figure 2: Wien-Bridge Oscillator Output Gain and Phase Shift

- It can be seen that at very low frequencies the phase angle between the input and output signals is “Positive” (Phase Advanced), while at very high frequencies the phase angle becomes “Negative” (Phase Delay). In the middle of these two points the circuit is at its

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 14

resonant frequency, (f_r) with the two signals being “in-phase” or 0° . Therefore this resonant frequency point can be defined with the following expression.

$$f_r = \frac{1}{2\pi RC}$$

Where:

f_r is the Resonant Frequency in Hertz

R is the Resistance in Ohms

C is the Capacitance in Farads

If we now place this RC network across a non-inverting amplifier which has a gain of $1+R_1/R_2$ the following basic Wien bridge oscillator circuit is produced.

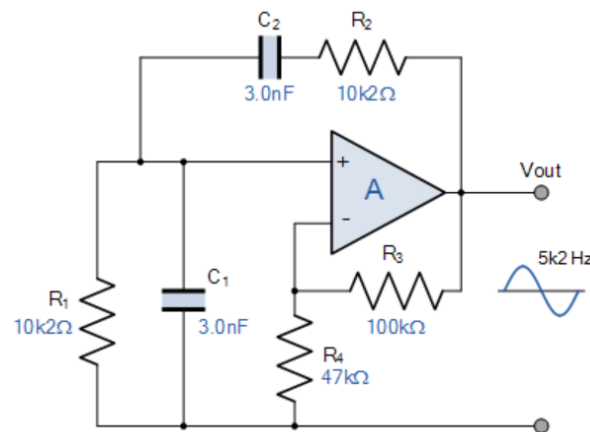


Figure 2: Wien-Bridge Oscillator by using op-amp

Procedure:

1. Make a circuit as shown in figure 2.
2. Connect CRO at output terminals and observe the wave form.
3. Calculate practically the frequency of oscillations by using the expression.

$$f = \frac{1}{T}, \text{ where } T = \text{number of space} \times \text{time/div}$$

4. Repeat the above steps 2, 3 by varying one parameter at a time that is, R, and C respectively and note down the practical values of oscillations of Wien-Bridge.
5. Compare the values of frequency of oscillations both theoretically and practically.
6. Also, calculate the gain(A).

Observation/Results:

R	C	Theoretical frequency (KHz)	Practical frequency (KHz)	AV

Mehran University of Engineering & Technology, Jamshoro
Department of Electronic Engineering
ES-203: Electronic Circuit Design
Lab Experiment No # 14

Review Questions:

- Write applications of Wien Bridge Oscillators?

- What are the limitations of Wien Bridge Oscillators?

- Compare and contrast RC Phase-shift and Wien-Bridge Oscillator?

- Why are LC circuits used more often than RC in demanding circuits?

Task:

Imitate the Wien-Bridge Oscillator by using transistor in multisim as shown below in figure 1. Observe its principal characteristics.